Specification and Estimation of Matrix Time Series Models with Multiple Terms

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Abstract

Matrix time series ($MaTS$) models allow to provide models for time series in the situation where for a number of regions the same set of variables for each region is observed. This leads to the observations at every time point being represented as a matrix. In the literature, matrix autoregressive models have been proposed that use one term per time lag. This imposes strong restrictions of generality that can be alleviated by adding terms. More terms on the other hand require identification restrictions. In this paper we propose such restrictions for the stationary and the integrated case. For co-integrated processes the cointegrating relations can be estimated super-consistently as usual. To determine cointegration relations, we propose a novel estimation method for models with higher order lags ($p > 1$) with the extension of Johansen's framework. With this notion, we investigate and introduce new critical values for trace statistics and extend them to the use of higher dimensions. Beside the estimation we also discuss specification of the integer parameters such as the number of time lags as well as the number of terms per time lag. We show that the familiar information criterion based model selection methods have properties as in the general multivariate case.

Keywords: cointegration rank, critical values, error correction model, estimation, matrix time series, trace test, specification

1. Introduction

Increasingly multivariate time series data contain more structure such that observations at a given time point can be arranged in a matrix leading to matrix time series ($MaTS$).[1] A prime example is constituted by observing the same variables in a number of regions. While such $MaTS$ can alternatively be viewed as multivariate time series by vectorising the matrices, this obscures the view to the structure of the time series: If columns refer to regions and rows to variables, then row operations correspond to linear combinations of variables for all countries while column operations relate to links of variables between regions.

In this paper, we hence consider $MaTS$, $(Y_t)_{t\in\mathbb{Z}}, Y_t \in \mathbb{R}^{M\times N}$ where M denotes the number of variables and N the number of regions.[2] The n-th column $Y_{t,:,n} \in \mathbb{R}^M$ then denotes the vector of observations for region n at time point t. Throughout upper case letters denote matrices (for example Y_t) and lower case letters the corresponding vectorized processes such that $(y_t)_{t\in\mathbb{Z}}, y_t \in \mathbb{R}^{MN}$, is the vector process obtained from columnwise vectorization of Y_t. We will also use ':' notation to indicate rows or columns of a matrix such that, for example, $Y_{t,m,:}$ denotes the m-th row and $Y_{t,:,n}$ the n-th column of the matrix Y_t.

In this notation the panel vector autoregressive model (see, e.g., the survey Canova and Ciccarelli [1]) often is formulated as

$$Y_t = A_1 Y_{t-1} + \cdots + A_p Y_{t-p} + U_t, \quad t \in \mathbb{Z}, \tag{1}$$

where $A_j \in \mathbb{R}^{M\times M}$ and $(U_t)_{t\in\mathbb{Z}}, U_t \in \mathbb{R}^{M\times N}$ is a white noise process. This definition implies that the same VAR model applies to each region:

$$Y_{t,:,n} = A_1 Y_{t-1,:,n} + \cdots + A_p Y_{t-p,:,n} + U_{t,:,n}, \quad t \in \mathbb{Z}.$$

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[1] In the literature, matrix time series also appear as matrix-valued or matrix-variate time series. We prefer to use the term 'matrix time series' because it encompasses all representations.

[2] This notation is used only for ease of presentation. The two dimensions not necessarily refer to variables and regions.

This model does not allow for spillovers wherein variables from one region are influenced by variables from other regions. Such influences can be included in the *matrix autoregressive model of order p (MAR(p))* as suggested in a series of papers, see, for example, Chen et al. [2, Remarks, p. 542], Li and Xiao [3], and Samadi and Billard [4] and the references therein):

$$Y_t = A_1 Y_{t-1} B_1' + \cdots + A_p Y_{t-p} B_p' + U_t, \quad t \in \mathbb{Z}. \tag{2}$$

This extension can, for example, be used to model error correction mechanisms showing a tendency to synchronize to the average across regions:

$$\Delta Y_t = -A_1 (Y_{t-1} - Y_{t-1} \frac{\iota_N}{N} \iota_N') + U_t = -A_1 Y_{t-1} (I_N - \frac{\iota_N}{N} \iota_N') + U_t \tag{3}$$

where $\iota_N = [1, ..., 1]' \in \mathbb{R}^N$ and $I_N \in \mathbb{R}^{N \times N}$ denotes the N dimensional identity matrix. Further, Δ denotes the time difference operator such that $\Delta Y_t = Y_t - Y_{t-1}$.

Such models have been studied intensively recently: Hecq et al. [5] added rank restrictions on the matrices A_j, B_j in the $MAR(p)$ model. Hsu et al. [6] developed a special $MAR(p)$ with rank restrictions and a banded structure matrix in a spatio-temporal setting. Tsay [7] extended the $MAR(p)$ to an autoregressive moving-average model ($MARMA$) by adding a moving average part. Zhang [8] extended the $MAR(1)$ model by adding additional terms in style of the multi-term model discussed below, wherein the additional terms have a restrictive structure. Lopetuso and Caporin [9] pointed out that using the $MAR(p)$ in an error correction framework is too restrictive to allow for an interpretation analogous to the VAR situation. This already shows in the simple co-integrating model (3), as the model defined in the MAR-type as an error correction model cannot be represented as a MAR in levels. This implies that the class of $MAR(p)$ models is not flexible enough in order to be useful for cointegration analysis when using both, a model in levels and the corresponding model in error correction formulation.

To mitigate these type of restrictions Li and Xiao [3] generalized the MAR model to the tensor autoregressive model with multiple terms and multiple lags (multi-term TenAR(p)) in the autoregression. When the order of the tensor is two, the model becomes a multi-term $MAR(p)$ model (abbreviated as $MAR_J(p)$) which is the focus of this paper:

$$Y_t = \sum_{j_1=1}^{J_1} A_{1,j_1} Y_{t-1} B_{1,j_1}' + \cdots + \sum_{j_p=1}^{J_p} A_{p,j_p} Y_{t-p} B_{p,j_p}' + U_t, \quad t \in \mathbb{Z} \tag{4}$$

where $A_{i,j_i} \in \mathbb{R}^{M \times M}$ and $B_{i,j_i} \in \mathbb{R}^{N \times N}$ for $i = 1, ..., p$. The model reduces to the $MAR(p)$ model for $J_i = 1$ for $\forall i = 1, ..., p$.

Vectorising this equation we obtain a $VAR(p)$ model for the vector process $(y_t)_{t \in \mathbb{Z}}$:

$$y_t = \left(\sum_{j_1=1}^{J_1} (B_{1,j_1} \otimes A_{1,j_1}) \right) y_{t-1} + \cdots + \left(\sum_{j_p=1}^{J_p} (B_{p,j_p} \otimes A_{p,j_p}) \right) y_{t-p} + u_t, \quad t \in \mathbb{Z}. \tag{5}$$

Noting that the number of elements in $\Phi_1 := \left(\sum_{j_1=1}^{J_1} (B_{1,j_1} \otimes A_{1,j_1}) \right)$ equals $J_1(N^2+M^2)$ whereas a full autoregressive matrix in the vectorised process includes $(MN)^2$ entries, it follows that the $MAR_{J_1}(1)$ for $J_1 < N^2M^2/(N^2 + M^2)$ contains less parameters than the corresponding $VAR(1)$.

Additionally the representation above is not identifiable even for $J_1 = 1$. Achieving identification is more complicated for $J_1 > 1$. Li and Xiao [3] and Hsu et al. [6] provide identifiability conditions based on the singular value decomposition (SVD) of a rearranged matrix $\mathcal{R}(\Phi_1)$, see below. The SVD is discontinuous at points with repeated singular values. This poses problems for uncertainty quantification also close to such points. It is not clear that these points are not of relevance for applications. Hence using the SVD for achieving identification does not appear to be optimal. Furthermore, the previous literature focused on the stationary case, while the co-integrated case is of great interest, for example, for macro-econometric or finance applications.

In this paper we consequently argue that the $MAR_J(p)$ model is more flexible and appropriate for the analysis of $MaTS$, in particular when modelling co-integration. The main contributions of this paper are to:

(I) investigate the unit root case of (co-)integrated processes, where we show that in this case multi-term models can be used effectively to avoid unrealistic assumptions imposed in single term models,

(II) introduce a different identification restriction related to the QR decomposition,

(III) provide the asymptotic properties of different estimators,

(IV) investigate the finite sample properties of model selection procedures (including specification of the cointegrating rank) for various sample sizes and dimensions of the matrix observations,

(V) calculate critical values for Johansen's type trace statistics for cointegration in matrix time series.

The rest of this paper is organized as follows. First, we introduce the identification restriction based on the QR decomposition. Then in Section 3 we discuss estimation and specification of the corresponding model in the stationary case. This is followed by a discussion of the error correction representation for the $MAR_J(p)$ model. The estimation of error correction for matrix autoregressive models ($vECMAR$) is discussed in Section 5. Section 6 provides simulation studies for stationary and co-integrated processes investigating the accuracy of the specification of integers as well as the prediction accuracy. Finally, Section 7 presents concluding remarks.

2. Identifiability

Identification and parameterisation for $VARs$ is simple: For given covariance sequence and known lag length p the VAR coefficients Φ_i are unique, each entry of them is a free parameter. The stability assumption leads to a convex open subset of $\mathbb{R}^{(NM)^2 p}$ as the parameter set.

For the $MAR_J(p)$ things are difficult: For instance, the pairs (A_{p,j_p}, B_{p,j_p}) and $(\frac{1}{c}A_{p,j_p}, cB_{p,j_p})$ provide the same Kronecker product $(B_{p,j_p} \otimes A_{p,j_p}), \forall p$. Constraining $\|A_{p,j_p}\| = 1, \forall p$, is not sufficient for identifiability, since $(B \otimes A) = ((-B) \otimes (-A))$ and $\|A\| = \|-A\|$. Thus other restrictions are needed to identify terms in the $MAR_J(p)$ representation of the $VAR(p)$ coefficients $\Phi_i = \sum_{j=1}^{J}(B_{i,j} \otimes A_{i,j})$.

The goal is to achieve an identifiable representation of the decomposition of Φ_i into a sum of Kronecker products, where the number of terms, J, is minimal and the matrices $A_{i,j}, B_{i,j}$ are unique and can be parameterized easily. Stated differently one searches for a mapping $\varphi : \theta \in \Theta \mapsto \mathbb{R}^{NM \times NM}$ such that $\varphi(\theta)$ is a bijective, continuously differentiable mapping with continuous inverse $\theta_i = \varphi^{-1}(\Phi_i), \Phi_i \in \varphi(\Theta)$, for a suitable parameter set $\Theta \subset \mathbb{R}^{d_\theta}$ where $d_\theta \leq J(MN)^2$ and a corresponding image set $\varphi(\Theta) \subset \mathbb{R}^{MN \times MN}, \overline{\varphi(\Theta)} = \Phi(M, N, J)$, the set of all sums of J Kronecker products of $M \times M$ with $N \times N$ matrices. The mapping φ typically is defined by finding restrictions for the matrices $A_{i,j}, B_{i,j}$ involved, such that the relation between Φ_i and the collection of matrices is bijective. The parameterization of the matrices then typically is straightforward.

Key to these restrictions is a rearrangement of the entries of the Kronecker products using the mapping $\mathcal{R}(.) : \mathbb{R}^{NM \times NM} \mapsto \mathbb{R}^{N^2 \times M^2}$ rearranging the elements of a matrix (see Van Loan and Pitsianis [10], Van Loan [11]).

Example 1. Consider the two matrices $A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$ and $B = \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix}$. Then

$$B \otimes A = \left(\begin{array}{cc|cc} b_{11}a_{11} & b_{11}a_{12} & b_{12}a_{11} & b_{12}a_{12} \\ b_{11}a_{21} & b_{11}a_{22} & b_{12}a_{21} & b_{12}a_{22} \\ \hline b_{21}a_{11} & b_{21}a_{12} & b_{22}a_{11} & b_{22}a_{12} \\ b_{21}a_{21} & b_{21}a_{22} & b_{22}a_{21} & b_{22}a_{22} \end{array}\right), \quad \mathcal{R}(B \otimes A) = \left(\begin{array}{c|c|c|c} b_{11}a_{11} & b_{21}a_{11} & b_{12}a_{11} & b_{22}a_{11} \\ b_{11}a_{21} & b_{21}a_{21} & b_{12}a_{21} & b_{22}a_{21} \\ b_{11}a_{12} & b_{21}a_{12} & b_{12}a_{12} & b_{22}a_{12} \\ b_{11}a_{22} & b_{21}a_{22} & b_{12}a_{22} & b_{22}a_{22} \end{array}\right) = \text{vec}(A)\text{vec}(B)'.$$

Comparing $B \otimes A$ and $\mathcal{R}(B \otimes A)$ one notices that both matrices contain the same elements but at different positions. Therefore, the mapping $\mathcal{R}$ can be seen as a bijective, linear mapping of $\mathbb{R}^{M^2 N^2}$ onto itself.

The important feature of $\mathcal{R}$ is that $\mathcal{R}$ converts the Kronecker product into the multiplication of the vectorisation of the matrices. Therefore, $\mathcal{R}(B \otimes A)$ has rank 1. □

The linearity of $\mathcal{R}$ implies that

$$\mathcal{R}(\Phi_p) = \mathcal{R}(\sum_{j_p=1}^{J_p}(B_{p,j_p} \otimes A_{p,j_p})) = \sum_{j_p=1}^{J_p} \text{vec}(A_{p,j_p})\text{vec}(B_{p,j_p})' = \mathcal{A}\mathcal{B}'$$

(where $\mathcal{A} \in \mathbb{R}^{M^2 \times J_p}, \mathcal{B} \in \mathbb{R}^{N^2 \times J_p}$) is of rank smaller or equal to J_p as the sum of J_p rank 1 matrices. Therefore, the mapping φ is obtained by identifying the factors $\mathcal{A}$ and $\mathcal{B}$ from the product $\mathcal{A}\mathcal{B}'$.

As proposed by Li and Xiao [3] (see also Tsay [7], Hsu et al. [6]) one possibility in this respect is the use of the singular value decomposition (SVD):

$$\mathcal{A}\mathcal{B}' = USV' = \begin{pmatrix} U_1 & \cdots & U_r \end{pmatrix} \begin{pmatrix} s_1 & 0 & \dots & 0 \\ 0 & s_2 & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 0 & \dots & 0 & s_r \end{pmatrix} \begin{pmatrix} V_1 & \cdots & V_r \end{pmatrix}, \tag{6}$$

where $U'U = I_r, V'V = I_r, 0 < s_1 \leq \cdots \leq s_r$. If the singular values s_j are distinct, the SVD is unique up to orientation changes for the singular vectors. Then $\mathrm{vec}(A_{p,j}) = U_j, \mathrm{vec}(B_{p,j}) = V_j s_j, j = 1, ..., r$ defines an identifiable parameterization where the sign can be fixed by assuming that the first entry in U_j is positive.

Assuming distinct singular values and non-zero entries for the first row of U_j is no real restriction, as this property holds for a generic subset of matrices $\mathcal{A}\mathcal{B}'$. However, numerical problems are expected also close to this set where, for example, s_i is close to s_{i+1}. One aspect in this regard is that the linearisation of the SVD required for the asymptotic normality for estimated singular values does only provide useful approximations for small samples, if $s_{i+1} - s_i$ is larger than four times the standard deviation such that confidence intervals for the two singular values do not overlap. Moreover, the linearisations involve the inverse of the distance $s_{i+1}^2 - s_i^2$ (see the Rayleigh-Schrödinger equation in Chapter 2 of Chatelin [12]).

Therefore, we prefer a different identification using the QR decomposition instead of the SVD. For this we need the definition of a positive upper triangular matrix (p.u.t.), compare Bauer and Wagner [13]:

Definition 1 (Positive upper triangular matrix). *A matrix $\mathcal{M} \in \mathbb{R}^{M \times N}, M \leq N$ having full row rank is said to be a positive upper triangular (p.u.t.) matrix if its n-th row $\mathcal{M}_n$ has the following form:*

$$\mathcal{M}_n = \begin{pmatrix} 0 & \cdots & 0 & \mathcal{M}_{n,m_n} & \cdots \end{pmatrix}, \mathcal{M}_{n,m_n} > 0$$

where $m_n > m_{n-1}, n = 1, ..., m$ (using $m_0 = 0$).

Therefore, in a p.u.t. matrix in each row the first non-zero entry is positive, and the next row has at least one more starting zero. The generic case for $N \geq M$ consists in $\mathcal{M} = [\mathcal{M}_u, \mathcal{M}_g], \mathcal{M}_u \in \mathbb{R}^{M \times M}$, which being upper triangular and invertible with only positive entries on the diagonal.

The significance of p.u.t. matrices stems from the fact that every matrix $\mathcal{R}(\Phi_i) \in \mathbb{R}^{M^2 \times N^2}$ of rank J_i can be uniquely factored as $\mathcal{R}(\Phi_i) = \mathcal{A}_i\mathcal{B}_i', \mathcal{A}_i \in \mathbb{R}^{M^2 \times J_i}, \mathcal{B}_i \in \mathbb{R}^{N^2 \times J_i}$, where $\mathcal{A}_i'\mathcal{A}_i = I_{J_i}$ and $\mathcal{B}_i'$ is p.u.t. Note, that $J_i \leq N^2$ such that $\mathcal{B}_i'$ is a wide matrix. The set of matrices $O_{M^2,J} := \{\mathcal{A}_i \in \mathbb{R}^{M^2 \times J}, \mathcal{A}_i'\mathcal{A}_i = I_J\}$ can be parameterised using Givens rotations as in Bauer et al. [14], see the next section.

For the QR decomposition, only one orthonormal matrix needs to be parameterised, while when using the SVD two such matrices are involved. We consider this to be a clear advantage.

Again, the non-generic case (where the heading $J_i \times J_i$ submatrix of $\mathcal{B}_i'$ is not of full rank) introduces discontinuities in the parameterisation. However, the p.u.t restriction can be alleviated by not choosing the heading square sub-matrix of $\mathcal{B}_i$ to be p.u.t. Any other square non-singular sub-matrix could be used. Since the matrix is of full column rank, such a sub-matrix must exist. This choice can also be done after estimation, possibly changing the restriction, if one finds a more suitable sub-matrix. Hence, the genericity assumption is not seen as a serious restriction. Contrary to the SVD case, no issues with close singular values arise, and there are no cross restrictions between parameters.

Sometimes we may want to include terms with more structure. Consider the following $MAR_2(1)$:

$$Y_t = A_{1,1}Y_{t-1}B_{1,1}' + Y_{t-1}B_{1,2}' + U_t$$

where, for example, $B_{1,2} = b_2\iota_N'/N$ implying a first order lagged term containing averages for all variables over all regions. This contains two terms where in the second term we restrict the $A_{1,2}$-matrix to be equal to the identity matrix. Consequently $\Phi_1 = B_{1,1} \otimes A_{1,1} + B_{1,2} \otimes I_M$. In this situation, we may orthogonalise such that $\mathrm{vec}(A_{1,1})'\mathrm{vec}(I_M) = 0$, that is, the vectorisation of $A_{1,1}$ lies in the orthogonal complement of the vectorisation of the identity matrix. This follows since (using $v_A := \text{vec}(A_{1,1})'\text{vec}(I_M)/M$)

$$B_{1,1} \otimes A_{1,1} + B_{1,2} \otimes I_M = B_{1,1} \otimes (A_{1,1} - v_A I_M) + (B_{1,2} + B_{1,1} v_A) \otimes I_M.$$

Then assuming that $v_A = 0$ leads to a unique decomposition. Note that for the first summand $\mathcal{R}(B_{1,1} \otimes (A_{1,1} - v_A I_M))$ has rank 1 in general. More generally, if $\mathcal{R}(\Phi_i)$ has rank J_i using the orthogonal complement to $\text{vec}(I_M)$ results in a $J_i - 1$ dimensional subspace that can be parameterised using Givens rotations (see Bauer et al. [14], Corollary 1).

In more detail, we use the following set of assumptions for the data generating process:

Assumption 1 (Data generating process)**.** *The vectorised matrix time series $(y_t)_{tz\in\mathbb{Z}}$ is assumed to be the stationary solution to equation (5) for $t \in \mathbb{Z}$ for independent identically distributed (i.i,d.) noise process $(u_t)_{t\in\mathbb{Z}}$ with expectation zero and variance $\Sigma > 0$ and finite fourth moments.*
Furthermore, with $\Phi_i := \left(\sum_{j_i=1}^{J_i}(B_{i,j_i} \otimes A_{i,j_i})\right)$ defining the matrix polynomial $A(z) = I_{MN} - \sum_{i=1}^{p} \Phi_i z^i$ it is assumed that $|A(z)| = 0$ implies that $|z| > 1$.
Finally, we assume that $\Phi_i \neq 0$ and for each $1 \leq i \leq p$ we have that $\mathcal{R}(\Phi_i) = \mathcal{A}_i \mathcal{B}_i'$ where $\mathcal{A}_i'\mathcal{A}_i = I_{J_i}$ and $\mathcal{B}_i'$ is p.u.t.

The assumptions state that $(y_t)_{t\in\mathbb{Z}}$ is a vector autoregressive process and that the integer valued parameters p and J_i, for $i = 1, ..., p$,, are correctly specified. The noise is assumed i.i.d. for simplicity. Alternatively, martingale difference sequence assumptions as in [15] would also be sufficient for our results below.

Finally, note that in the special case of $J_i = 1$, for $i = 1, ..., p$, the identification restriction implies $\mathcal{A}_i'\mathcal{A}_i = 1$ equivalent to the Frobenius norm of A_i being equal to 1 and the (1,1) element of $B_{1,1}$ being positive.

Note that here we do not impose any restriction on the noise covariance matrix $\Sigma > 0$ (except for non-singularity). Often, noise U_t is assumed to follow a *separable* multivariate normal distribution in the sense that $\mathbb{V}(u_t) = \Sigma_c \otimes \Sigma_r$ (see, e.g., Chen et al. [2], Samadi and Billard [4], Gupta and Nagar [16]). The Kronecker structure for the variance matrix implies that the correlation structure within each region (that is, for $U_{t,:,n}, n = 1, ..., N$) is identical as is the correlation structure for the vector of all regional observations of a given variable (that is, $U_{t,m,:}, m = 1, ..., M$). This can be used for an estimation. In the following, we will point out when the separability is assumed.

3. Estimation and Specification in the Stationary Case

The bilinear nature of the $MAR_J(p)$ model complicates the estimation of parameters. We discuss alternating methods and direct optimization of the Gaussian likelihood for parameter estimation. Consequently, a number of numerical algorithms are available that aim at optimizing the Gaussian likelihood. They can be seen as ways in order to obtain initial estimates for subsequent nonlinear optimization.

3.1. Alternating optimization

Recall the defining equation for the $MAR_J(p)$ model:

$$Y_t = \sum_{i=1}^{p} \sum_{j=1}^{J_i} A_{i,j} Y_{t-i} B_{i,j}' + U_t = \boldsymbol{A}^* X_t \boldsymbol{B}^{*\prime} + U_t \tag{7}$$

where $\boldsymbol{A}^* = [A_{1,1}, ..., A_{p,j_p}]$ and $\boldsymbol{B}^* = [B_{1,1}, ..., B_{p,j_p}]$ are $M \times M(J_1 + ... + J_p)$ and $N \times N(J_1 + ... + J_p)$ coefficient matrices respectively. Furthermore,

$$X_t = diag[Y_{t-1}, \cdots, Y_{t-1}, Y_{t-2}, \cdots, Y_{t-2}, \cdots, Y_{t-p}, \cdots, Y_{t-p}] \in \mathbb{R}^{M(J_1+...+J_p)\times N(J_1+...+J_p)}.$$

Given the integers $p, J_1, ..., J_p$ one notices that the equations are linear in $\boldsymbol{A}^*$ given $\boldsymbol{B}^*$ and vise versa. This can be used for an alternating optimization (AO) procedure as described in the Algorithm 1. Mathematical background can be found in the supplementary material.

This algorithm improves the Gaussian likelihood in each step and hence converges to a local optimum. It requires an initial guess for either $\boldsymbol{A}^*$ or $\boldsymbol{B}^*$. A simple idea to obtain such an initial guess is to use the VAR(p) estimates $\hat{\Phi}_i \in \mathbb{R}^{MN\times MN}$ for the vectorized system. Although these estimates will not have the structure of a sum of J_i Kronecker

Algorithm 1 Alternating optimization with general covariance matrix.

Require: $\widehat{\boldsymbol{A}^*}_{(0)}$
$i \leftarrow 1$
while not converged **do**
Calculate $\hat{X}_{t,(i)} \leftarrow \widehat{\boldsymbol{A}^*}_{(i)} X_t \in \mathbb{R}^{M\times N(J_1+...+J_p)}$
Regress Y_t onto $\hat{X}_{t,(i)}$ to obtain $\widetilde{\boldsymbol{B}^*}_{(i)}$
Calculate $\tilde{X}_{t,(i)} \leftarrow X_t \widetilde{\boldsymbol{B}^*}_{(i)}' \in \mathbb{R}^{M(J_1+...+J_p)\times N}$
Regress Y_t onto $\tilde{X}_{t,(i)}$ to obtain $\widetilde{\boldsymbol{A}^*}_{(i)}$ and $\hat{\Sigma}$ (empirical variance of vectorised residuals).
Calculate $\hat{\Phi}_{j,(i)}$, $j = 1, ..., p$ corresponding to $\widetilde{\boldsymbol{A}^*}_{(i)}, \widetilde{\boldsymbol{B}^*}_{(i)}$.
$i \leftarrow i + 1$
Refactor $\mathcal{R}(\hat{\Phi}_{j,(i-1)})$ to obtain $\widehat{\boldsymbol{A}^*}_{(i)}, \widehat{\boldsymbol{B}^*}_{(i)}$ fulfilling the identification conditions.
end while

products, finding the closest sum of Kronecker products is simple; see the Algorithm 2. Since the reordering operator $\mathcal{R}$ only permutes the entries, the approximation error in terms of the Frobenius norm $\|\hat{\Phi}_i - \tilde{\Phi}_i\|_{Fr}$ is determined by the sum of the omitted singular values.

If separability of the variance is assumed, regressions can be adjusted for correlations in rows and/or columns, resulting in maximum likelihood estimation (MLEs) under the matrix normal distribution, i.e., $U_t \sim \mathcal{N}_{M,N}(0, \Sigma_c \otimes \Sigma_r)$ as shown in Algorithm S1, which also provides an estimate of the separable variance. (Mathematical background of MLE estimators and Algorithm are given in the Supplement S1.)

Since the unrestricted estimator $\hat{\Phi}_i$ in the VAR is consistent, the same holds for the nearest sum of the Kronecker products approximation, provided the specification is correct. Factorization using QR decomposition is a continuous operation in the generic situation. This shows consistency for the initial values. Consistency for the AO iterations then follows, as these increase the likelihood.

Algorithm 2 Nearest sum of Kronecker products

Require: $\hat{\Phi}$
Require: J
Calculate $\hat{\mathcal{R}} \leftarrow \mathcal{R}(\hat{\Phi})$
Calculate the SVD $\hat{\mathcal{R}} = \hat{U}\hat{S}\hat{V}'$ where the diagonal values in $\hat{S}$ are ordered decreasing in size.
Approximate $\hat{\mathcal{R}}_J = \hat{U}_J \hat{S}_J \hat{V}_J'$ where $\hat{U}_J \in \mathbb{R}^{M^2\times J}, \hat{V}_J \in \mathbb{R}^{N^2\times J}$.
$\hat{\mathcal{A}} \leftarrow \hat{U}_J, \hat{\mathcal{B}} \leftarrow \hat{V}_J \hat{S}_J$
$\tilde{\Phi} \leftarrow \mathcal{R}^{-1}(\hat{\mathcal{A}}\hat{\mathcal{B}}')$

3.2. *Optimizing the Gaussian likelihood*

As alternating optimization does not guarantee convergence in finite samples (similar to the EM algorithm), direct optimization of the Gaussian likelihood is an option. In this way, additional restrictions (such as stability of the VAR polynomial) can also be imposed.

The Gaussian likelihood uses the residuals

$$\hat{U}_t(\theta) = Y_t - \boldsymbol{A}^*(\theta) X_t \boldsymbol{B}^*(\theta)' \tag{8}$$

where the notation $\boldsymbol{A}^*(\theta)$ and $\boldsymbol{B}^*(\theta)$ stresses the dependence on an underlying parameter vector θ which also parameterizes the variance $\Sigma(\theta)$. Then $-2/T$ times the logarithm of the Gaussian likelihood equals (up to constants; conditional on the first p observations):

$$ll_T(\theta) = \log|\Sigma_u(\theta)| + \mathrm{tr}[\frac{1}{T}\sum_{t=p+1}^{T} \hat{u}_t(\theta)\hat{u}_t(\theta)'\Sigma_u(\theta)^{-1}]. \tag{9}$$

Here $\hat{u}_t(\theta) = \text{vec}(\hat{U}_t(\theta))$. If $\Sigma(\theta)$ is unrestricted, then

$$ll_T(\hat{\theta}) = \log|\hat{\Sigma}_u(\hat{\theta})| + MN, \qquad \hat{\Sigma}_u(\hat{\theta}) = \frac{1}{T}\sum_{t=p+1}^{T} \hat{u}_t(\theta)\hat{u}_t(\theta)'. \tag{10}$$

To this end, note that the identification restriction can be used to provide a parameterization of the set of all systems under the genericity assumption that the heading $(J_i \times J_i)$ sub-block of $\mathcal{B}_i'$ is nonsingular for all $i = 1, ..., p$:

Definition 2 (Parameterization)**.** *The parameter vector $\theta = (\theta_1', ..., \theta_p', \sigma')' \in \Theta \subset \mathbb{R}^d$ is given by the following components for $i = 1, ..., p$ separately: For i such that $J_i < M^2$ the parameter vector $\theta_i \in \mathbb{R}^{d_i}$ is composed of:*

- *The J_i positive diagonal entries of $\mathcal{B}_i'$ in $\mathbb{R}_+ = (0, \infty)$*
- *The remaining $J_i N^2 - J_i(J_i + 1)/2$ free entries of $\mathcal{B}_i'$ in $\mathbb{R}$.*
- *$M^2 J_i - J_i(J_i + 1)/2$ parameters for the set $O_{M^2,J_i} := \{U_i \in \mathbb{R}^{M^2 \times J_i}, U_i' U_i = I_{J_i}\}$ (see Lemma 1 in Bauer et al. [14]).*

Consequently, then $d_i = J_i(M^2 + N^2) - J_i^2$.

For i such that $J_i = M^2$ then $\mathcal{A}_i = I_{M^2}$ and the parameters consist in all entries of B_{i,j_i} such that $d_i = M^2 N^2$.

The covariance matrix $\Sigma = LL' \in \mathbb{R}^{MN \times MN}$ can be parameterized using the Cholesky factors L such that $\sigma \in \mathbb{R}^{MN(MN-1)/2} \times \mathbb{R}_+^{MN}$ (restricting the diagonal entries in the Cholesky factor to be positive).

Then $d = \sum_{i=1}^{p} d_i + MN(MN + 1)/2$.

The parameterization then is given by the function $\varphi : \Theta \to \{A_{i,j}, B_{i,j}, i = 1, ..., p, j = 1, ..., J_i, A_{i,j} \in \mathbb{R}^{M \times M}, B_{i,j} \in \mathbb{R}^{N \times N}\}$ such that $\varphi(\theta) = (A_{i,j}(\theta), B_{i,j}(\theta))$.

It follows from Bauer et al. [14] Lemma 1 that the mapping φ for given integers $(p, J_1, ..., J_p)$ is continuously differentiable on Θ with continuously differentiable inverse on $\Lambda := \varphi(\Theta)$. The points of discontinuity relate either to entries in $\mathcal{B}_i$ restricted in the p.u.t. form being zero or discontinuities of the parameterization of the orthonormal columns. These are parameterized using Givens rotations, which are based on 2×2 matrices rotating the coordinate systems by an angle $\omega \in [0, 2\pi)$. The dependence on the angle is discontinuous at the angle $\omega = 0$, which coincides with $\omega = 2\pi$. Clearly, such points of discontinuity form a null set such that the complementary set is open and dense in $O_{M^2,J}$, compare [14] Lemma 1 and Remark 13.

With this parameterization the asymptotic properties of the quasi maximum likelihood estimator can be given (the proof can be found in the Appendix B):

Theorem 1. *Let the process $(Y_t)_{t \in \mathbb{Z}}$ be generated according to Assumption 1 corresponding to the true parameter vector $\theta_\circ$, which is assumed to be a point of continuity of φ and an interior point of Θ. Assume that the indices $(p, J_1, ..., J_p)$ are known and used for estimation. The estimator $\hat{\theta}$ maximising the Gaussian likelihood (9) has the following properties:*

(i) Consistency: $\hat{\theta} \to \theta_\circ$.

(ii) Asymptotic normality: $\sqrt{T}(\hat{\theta} - \theta_\circ) \xrightarrow{d} \mathcal{N}(0, V)$.

Thus, the asymptotics in the stationary case are straightforward and do not deviate from the usual settings. Specification tests can be performed using likelihood ratio tests, which have standard χ^2 asymptotic distributions under the null hypothesis.

In order to specify the integers p and $J_1, ..., J_p$ alternatively information criteria like the Akaike information criterion (AIC), the Schwarz information criterion (SBIC), and the Hannan-Quinn information criterion (HQC), can be used (see, for example, Hannan and Deistler [15]), where $\mathcal{I}_p = (p, J_1, ..., J_p)$ denotes the multi-index:

$$AIC(\mathcal{I}_p) = log|\hat{\Sigma}_u(\mathcal{I}_p)| + \frac{2}{T}\vartheta(\mathcal{I}_p).$$

$$SBIC(\mathcal{I}_p) = log|\hat{\Sigma}_u(\mathcal{I}_p)| + \frac{log(T)}{T}\vartheta(\mathcal{I}_p).$$

$$HQC(\mathcal{I}_p) = log|\hat{\Sigma}_u(\mathcal{I}_p)| + \frac{2log(log(T))}{T}\vartheta(\mathcal{I}_p).$$

where $\hat{\Sigma}_u(\mathcal{I}_p) = \hat{\Sigma}_u(\hat{\theta}(\mathcal{I}_p))$ for $\theta(\mathcal{I}_p)$ denoting the estimated parameter vector for integer indices specified as $\mathcal{I}_p$. Then the selected lag order and term number can be obtained based on the minimum value derived from AIC, SBIC and HQC. By selecting J_i we also decide between $MAR(p)$ (for $J_i = 1, \forall i$), $VAR(p)$ (for $J_i = \min(M^2, N^2)$) or $MAR_J(p)$.

4. Co-Integration and error correction representation

The above description contains only the stationary case. In Li and Xiao [17] and Hecq et al. [18], the integrated case is also considered by proposing the Matrix Error Correction Model ($MECM$) for the $MAR(p)$ model. This extends the theory from the VAR framework for the vectorized process to the $MaTS$ setting. However, $MAR(p)$ – being a heavily restricted VAR – introduces some unwanted limitations leading to differences when working with the model in levels compared to working with error correction formulations.

These differences can already be seen in the $MAR(1)$ case of only one lag being present. The corresponding vectorization then is a $VAR(1)$:

$$y_t = \Phi_1 y_{t-1} + u_t.$$

Assuming that no explosive roots exist ($\lambda_{|max|}(\Phi_1) \leq 1$) we may find integration either in levels or in error correction:

1. Model in levels: Eigenvalues of Φ_1 equal to 1 correspond to integration, co-integration amounts to eigenvalues of Φ_1 smaller than one in modulus. For a diagonalisable matrix Φ_1 this corresponds to $\Phi_1 \neq I_{MN}$.
2. Error correction: The vector error correction model ($VECM$) for one lag amounts to $\Delta y_t = y_t - y_{t-1} = \Pi y_{t-1} + u_t$ where $\Pi = \Phi_1 - I_{MN}$. Integration occurs for singular matrix Π, while co-integration appears if additionally $\Pi \neq 0$.

Thus, the multiplicity of the eigenvalue 1 of Φ_1 or equivalently the rank of $\Phi_1 - I_{MN}$ determines the properties of the process $(y_t)_{t\in\mathbb{Z}}$.

For the $MaTS$ case things are more complicated. Co-integration has been defined in Li and Xiao [17] using the equation for $J_1 = 1$:

$$Y_t = A_{1,1} Y_{t-1} B_{1,1}' + U_t \tag{11}$$

Mimicking the $VECM$ we could also define co-integration using an error correction formulation:

$$\Delta Y_t = \alpha_r \beta_r' Y_{t-1} \beta_c \alpha_c' + U_t. \tag{12}$$

These two equations have different implications for the number of integrated and stationary variables as well as on cointegrating relations:

Theorem 2. *(I) Let $(Y_t)_{t\in\mathbb{Z}}$ be defined via (11) where all eigenvalues of $B_{1,1}$ and $A_{1,1}$ are inside the open unit disc or at $z = 1$. Let the multiplicity of the eigenvalue $z = 1$ of $A_{1,1}$ equal n_A where $A_{1,1} = T_A diag(I_{n_A}, A_{1,1,\bullet})T_A^{-1}$. Here $A_{1,1,\bullet}$ is stable with all eigenvalues smaller than one in modulus. Analogously, let the multiplicity of the eigenvalue $z = 1$ of $B_{1,1}$ be denoted as n_B, where $B_{1,1} = T_B diag(I_{n_B}, B_{1,1,\bullet})T_B^{-1}$.*
Then the vectorized process $(y_t)_{t\in\mathbb{Z}}$ has $n_A n_B$ common trends, the remaining components being stationary. In particular in $T_A^{-1} Y_t (T_B^{-1})'$ the heading $n_A \times n_B$ submatrix contains the common trends, all remaining entries being stationary.
(II) Let $(Y_t)_{t\in\mathbb{Z}}$ be generated according to (12) where $\alpha_r \beta_r' \in \mathbb{R}^{M\times M}$ has rank r_r and where $\alpha_c \beta_c' \in \mathbb{R}^{N\times N}$ has rank r_c. Assume that the matrix polynomial $A(z) = \Delta(z) - (\alpha_c \beta_c' \otimes \alpha_r \beta_r')z$ has all its roots outside the unit circle or at $z = 1$.
Then the vectorised process $(y_t)_{t\in\mathbb{Z}}$ has $r_r r_c$ stationary components and hence $MN - r_r r_c$ common trends. The stationary co-integrating relations are given as $\beta_r' Y_t \beta_c$.

In the general cases of a $MAR(p)$ process for $p > 1$ the common trends are found from the zeros of the autoregressive polynomial $A(z)$ evaluated at $z = 1$, i.e. $A(1) = I_{MN} - \sum_{j=1}^{p}(B_j \otimes A_j)$. This is the sum of p Kronecker products, and hence it is not necessarily a Kronecker product itself.

If the error correction formulation is used to define the model, then generalizations are possible, since then the increase in the number of lags leaves the specification of Π unchanged. Then, for every intra-regional cointegrating relation also simultaneously cross-region relations need to be applied in order to obtain stationary processes. Therefore, intra-regional cointegrating relations may apply to no country or only to a set of countries.

As an aside, note that the cointegrating rank for the error correction case is not unrestricted but needs to equal $r_r r_c$, the product of two integers.

Additionally, the two model classes ($MAR(p)$ in levels and $MECM$) define different subsets of the $VAR(p)$ processes for the vectorized process. This can be seen by rewriting $MAR(3)$ in levels into an error correction formulation:

$$\begin{aligned} Y_t &= A_1 Y_{t-1} B_1' + A_2 Y_{t-2} B_2' + A_3 Y_{t-3} B_3' + U_t, \\ \Rightarrow \Delta Y_t &= Y_t - Y_{t-1} = A_1 Y_{t-1} B_1' - Y_{t-1} + A_2 Y_{t-2} B_2' + A_3 Y_{t-3} B_3' + U_t \\ &= A_1 Y_{t-1} B_1' - Y_{t-1} + A_2 Y_{t-2} B_2' + A_3 Y_{t-2} B_3' - A_3 \Delta Y_{t-2} B_3' + U_t. \\ &= A_1 Y_{t-1} B_1' - Y_{t-1} + A_2 Y_{t-1} B_2' + A_3 Y_{t-1} B_3' - A_2 \Delta Y_{t-1} B_2' - A_3 \Delta Y_{t-1} B_3' - A_3 \Delta Y_{t-2} B_3' + U_t. \end{aligned}$$

This shows that a $MAR(p)$ model in levels corresponds to a multi-term model if written in error correction form. This is a direct consequence of the summation of terms when building the error correction representation. Note that this also holds for the Y_{t-1} term, which in the above example of $MAR(3)$ sums up four Kronecker product terms in vectorization.

In general, the error correction model in the $MAR(p)$ form for the level becomes a special form of multi-term $MAR(p)$ model in the error correction formulation:

$$\Delta Y_t = -Y_{t-1} + \sum_{i=1}^{p} A_i Y_{t-1} B_i' - \sum_{k=1}^{p-1} \sum_{i=k+1}^{p} A_i \Delta Y_{t-k} B_i' + U_t \tag{13}$$

Note that in vectorisation the term $\sum_{i=k+1}^{p} B_i \otimes A_i$ may be representable by less than $p-k$ terms, although generically it will not. This holds in particular for the term involving Y_{t-1} tied to the cointegrating relations:

$$\alpha\beta' = \Pi = -I_{MN} + \sum_{i=1}^{p}(B_i \otimes A_i) \tag{14}$$

Even for $p = 1$ this is the sum of two Kronecker product terms. Correspondingly, the matrices α and β typically do not have a Kronecker product structure. This makes interpretation in the $MaTS$ sense difficult.

Analogously, a model defined as a one-term model in the error correction formulation corresponds to a multi-term model in levels:

$$\begin{aligned} \Delta Y_t &= \Pi_r Y_{t-1} \Pi_c' + A_1 \Delta Y_{t-1} B_1' + A_2 \Delta Y_{t-2} B_2' + U_t \\ &= \Pi_r Y_{t-1} \Pi_c' + A_1 Y_{t-1} B_1' - A_1 Y_{t-2} B_1' + A_2 Y_{t-2} B_2' - A_2 Y_{t-3} B_2' + U_t. \end{aligned}$$

This model has one term in the last lag and at most two terms for all other lags and three terms for the first lag (taking Y_{t-1} from the left hand side into account).

As an alternative, we propose to use an unrestricted formulation for the matrix Π corresponding to the vectorized equation in combination with a multi-term formulation ($vECMAR_J(p-1)$ model):

$$\Delta y_t = \alpha\beta' y_{t-1} + \sum_{k=1}^{p-1} \sum_{i=1}^{J_k} (\Gamma_{c,k,i} \otimes \Gamma_{r,k,i}) \Delta y_{t-k} + u_t \tag{15}$$

Here $\alpha, \beta \in \mathbb{R}^{MN \times r}$ are matrices without explicitly imposing the Kronecker product structure. The multi-term structure for the differenced terms may be restricted to correspond to the $MAR(p)$ in levels. Alternatively, we may specify the number of terms independently directly in the error correction formulation.

In addition, deterministic terms $\Psi_0 + \Psi_1 t$ can be added following the five different specifications of Johansen [19] as can be other additional deterministic terms.

5. Estimation and Specification in the I(1) Case

Estimation of $vECMAR_J(p-1)$ (15) can be achieved by quasi-MLE, optimising the Gaussian likelihood. Identification for multi-terms, as well as the product $\alpha\beta'$ can be achieved using QR-decomposition as described above. Hereby $\beta'\beta = I_r$ and α' being p.u.t. is used leading to a parameter vector θ_β corresponding to $\beta(\theta_\beta)$, see Bauer et al. [14] Lemma 1. From this representation we can see that part of the parameter vector called $\theta_{\beta,L}$ determines the column space, while the remaining parameters $\theta_{\beta,R}$ rotate the basis of the column space.

This leads to the definition of an identified parameter vector $\theta \in \Theta \subset \mathbb{R}^d$ for a suitable dimension d. Restrictions such as separability for Σ or the usage of a $MAR(p)$ model in levels can easily be incorporated.

The asymptotic properties of the corresponding estimators are readily obtained since $vECMAR$ is essentially a restricted $VECM$ model:

Theorem 3. *Let the data be generated according to* (15)*, where the noise $(u_t)_{t\in\mathbb{Z}}$ is (i.i.d.) with expectation zero, variance $\Sigma \in \mathbb{R}^{MN\times MN}, \Sigma > 0$, and finite fourth moments.*
The lag polynomial corresponding to (15) *has no roots inside the unit circle or on the unit circle except for $z = 1$. The matrix $\Pi = \alpha\beta'$ has rank r.*
Let $\theta_\circ \in \Theta \subset \mathbb{R}^d$ denote the parameter vector for the model, where $\theta_\circ$ is an interior point of Θ.
Then let $\hat{\theta}$ denote the estimator maximising the Gaussian likelihood over Θ based on a data set of sample size T.

(I) Then $\hat{\theta} \xrightarrow{p} \theta_\circ$, that is, the estimator is (weakly) consistent.

(II) With $\hat{\beta}$ denoting the estimator for β, we have $T^\gamma d_{gap}(\hat{\beta}, \beta_0) \xrightarrow{p} 0$ for all $0 < \gamma < 1$, that is, $\hat{\beta}$ is estimated super consistently. Here d_{gap} denotes the gap metric for the distance of vector spaces (see, e.g., Chatelin [12]).

(III) Denoting with $\theta_{\beta,L}$ the part of θ corresponding to the column space of β, the asymptotic distribution of $T(\hat{\theta}_{\beta,L} - \theta_{0,\beta,L})$ is mixed normal, while the remaining parameter vector is asymptotically normally distributed.

This result provides the usual asymptotic properties of the various parts of the parameter vector.

5.1. Estimation

Since the parameter vector for the $vECMAR_J(p-1)$ model is typically large, it is advantageous to use the alternating optimization idea: The model equation (15) shows that the bilinear part involves the differenced terms. Analogously to (7) we have:

$$\Delta y_t = \alpha\beta' y_{t-1} + \mathrm{vec}(\mathcal{A}_\Delta \Delta X_{t-1} \mathcal{B}_\Delta') + u_t. \tag{16}$$

For given $\Gamma_{c,k,i}$'s (implying a given $\mathcal{B}_\Delta$) we obtain a standard (but restricted) $VECM$ model. Analogously for given $\mathcal{A}_\Delta$. [3] Hence we can iterate as in Algorithm 1, combining with reduced rank regression (see, Anderson [20, 21]) in the classical vector cointegration analysis Johansen [19].

Alternatively, we can acknowledge that for known $\alpha\beta'$ we obtain the matrix equation

$$\Delta Y_t - \mathrm{Mat}(\alpha\beta' y_{t-1}) = \mathcal{A}_\Delta \Delta X_{t-1} \mathcal{B}_\Delta' + u_t \tag{17}$$

where Mat(.) denotes the matrisation operator (inverse of vectorization). These two equations suggest a different AO scheme iterating between the $MAR_J(p)$ for $\Delta Y_t - \mathrm{Mat}(\alpha\beta' y_{t-1})$ and the error correction part for $\Delta y_t - \mathrm{vec}(\mathcal{A}_\Delta \Delta X_{t-1} \mathcal{B}_\Delta')$. This is detailed in Algorithm 3. The advantage of this algorithm is that it incorporates cointegration rank specification while imposing restrictions on the coefficients. Moreover, the fulfillment of the stability condition of the system can be checked. Violation of the condition is then considered as an indication of insufficient flexibility of the model and/or an inappropriate cointegration rank.

[3] Here, $\mathcal{A}_\Delta$, $\mathcal{B}_\Delta$ and ΔX_{t-1} are defined as in Equation (7).

5.2. *Specification of the cointegrating rank*

Relaxing restrictions on the matrix Π provides an advantage in terms of the specification of the cointegrating rank, as the usual trace test in the $VECM$ framework can be easily adapted. In the notation of Johansen [19] the model is written as

$$z_{0t} = \alpha\beta' z_{1t} + \Psi_2 z_{2t} + u_t \tag{18}$$

where $z_{0t} = \Delta y_t, z_{1t} = y_{t-1}$ and z_{2t} subsumes all lagged differences as well as deterministic terms (which may also be contained in z_{1t} implementing restrictions). Then the first step of the Johansen's procedure is to eliminate the impact of z_{2t} from the right hand side. This results in

$$r_{0t} = \alpha\beta' r_{1t} + \tilde{u}_t.$$

This equation features the rank restriction which can be included in the estimation using eigenvalue calculations. These involve matrices $S_{ij} = T^{-1} \sum_{t=p+1}^{T} r_{it} r'_{jt}, i, j \in \{0, 1\}$. In the rank restricted regression, the maximal likelihood values for different values of r can be calculated based on only one eigenvalue decomposition, and hence numerically efficiently.

Algorithm 3 $vECMAR_J(p-1)$ estimation with determination of the cointegration rank.

Require: Number of lags $p-1$
Require: Initial estimate $MAR_J(p)$ s.t. $J = [j_1, j_2..., j_p] = [p+1, p-1, p-2, \ldots, 1]$
  $l \leftarrow 0$
  while estimate not marginally stable **do**
    Initial start $\hat{r}_{0,t,(l)} \leftarrow MAR_J(p) \quad s.t. \quad rank_0 = l$
    $\hat{r}_{0,t,(i)} \leftarrow \hat{r}_{0,t,(l)}$
    $i \leftarrow 1$
    while not converged **do**
      Rank restricted regression $\hat{\alpha}_{(i)}, \hat{\beta}_{(i)}, \widehat{rank}_i \leftarrow \hat{r}_{0,t,(i)} = \alpha\beta' y_{t-1} + \hat{u}_{t,(i)}$ using the trace test
      Calculate $\hat{r}_{t,(i)} \leftarrow \Delta y_t - \hat{\alpha}_{(i)}\hat{\beta}'_{(i)} y_{t-1}$
      Matricize $\hat{R}_{t,(i)} \leftarrow \mathrm{Mat}(\hat{r}_{t,(i)})$
      Estimate the $ExoMAR_J(p)$ $\hat{\mathcal{A}}_{\Delta,i}, \hat{\mathcal{B}}_{\Delta,i} \leftarrow \hat{R}_{t,(i)} = \mathcal{A}_\Delta \Delta X_{t-1} \mathcal{B}'_\Delta + \hat{u}_t$
      Calculate $\hat{R}_{0,t,(i)} \leftarrow \Delta Y_t - \hat{\mathcal{A}}_{\Delta,i} \Delta X_{t-1} \hat{\mathcal{B}}'_{\Delta,i}$
      Vectorize $\hat{r}_{0,t,(i)} \leftarrow \mathrm{vec}(\hat{R}_{0,t,(i)})$
      Calculate residuals $\hat{u}_{t,(i)}$
      $i \leftarrow i + 1$
    end while
    Check the stability condition for implied $vECMAR$
    if Stability condition Satisfied **then**
      Exit
    else
      $l \leftarrow l + 1$
      if $l >$ number of dimensions **then**
        Exit with no stable solution
      end if
    end if
  end while

The asymptotic properties of the trace test then rely on the asymptotics of these quantities. These are presented, for example, in Lemma 10.3 of Johansen [19]. From inspecting these results, it follows that the concentration step is not essential to eliminate the term $\Psi_2 z_{2t}$ on the right hand side. This could also be achieved by subtracting $\hat{\Psi}_2 z_{2t}$ from the equation resulting in $r_{0t} = \Delta y_t - \hat{\Psi}_2 z_{2t}$ for a consistent $\hat{\Psi}_2 \to \Psi_2$. This is done in Algorithm 3 for the calculation of $\hat{r}_{0,t,(i)}$. Consistency of the estimates is sufficient for the usual asymptotics for the trace test statistic. Since the AO procedure for correctly specified J maximizes the Gaussian likelihood, consistency follows from standard arguments as for Theorem 3.

5.3. *Critical values for cointgeration estimation*

Determining the cointegration rank requires the use of trace test statistics, as the estimation method discussed above, along with its associated Algorithm 3, repeatedly applies the trace test to determine the cointegration rank and find the optimum. For Johansen's type of cointegration test, related tables of critical values are widely available in the literature (i.e. Johansen [19], MacKinnon et al. [22]). Additionally, many econometric software packages use critical values from different sources, such as Johansen [19], MacKinnon et al. [22], Osterwald-Lenum [23]. [4] For certain tests or situations, these packages may only provide asymptotic critical values, which are based on large sample sizes. However, in real-world applications—where data availability is often limited—it is important to interpret these results with caution, especially when working with small samples. Moreover, several studies have noted discrepancies between the finite-sample and asymptotic properties of the trace test. Johansen [24], Turner [25] Consequently, the literature contains varying critical values depending on the author and whether an approximation methodology is used.

Small sample sizes require adjusted critical values that differ from asymptotic values to account for reduced test power, so standard asymptotic tables are inadequate in such cases, and specialized tables or simulation-based critical values should be used instead. These adjustments should reflect the specific sample size (T) and the number of variables (K) in the model.

In addition to the sample size (T), the distribution of the likelihood ratio test statistics also depends on the parameters (θ) under the null hypothesis. When the data-generating process (DGP) incorporates a $MaTS$ structure, restrictions are imposed on the parameter space. While this dependence on θ disappears, so do the imposed restrictions, as $T \to \infty$ (removing the effect of imposed restrictions asymptotically), in finite samples, such restrictions do affect the distribution. This discrepancy may lead to misleading results when comparing models or conducting empirical studies. To address this issue, this paper provides both asymptotic and size adjusted critical values (for $T = 200, 500$) for the DGP associated with the model comparisons presented herein. Moreover, the critical values are extended to higher dimensions, as the existing tables in the literature are limited. For example, MacKinnon et al. [22] widely used extensive Monte Carlo-based critical values, which are generally accepted in the literature, cover up to 12 dimensions ($K = 12$). So, this paper computes and extends critical values up to 20 dimensions, corresponding to the (4×5) dimensional $MaTS$ process.

Before concluding this section, it is worth noting that the tables provided in this paper include only the critical values used in the simulation experiments for model comparison. Critical value tables that incorporate deterministic components—such as constant terms or linear trends, which are part of Johansen's five deterministic case framework—are beyond the scope of this paper, although such extensions can be incorporated into our approach and the associated algorithms. We leave this line of inquiry, as well as more extensive simulation studies for high dimensional settings, for future research. The approximated critical values are presented in Table A.2 with additional simulation details in Appendix A and in Table-S1 of Supplement S3.

5.4. *Specification of integer valued parameters*

The $vECMAR_J(p-1)$ model has a great number of integer valued parameters such that the lag length $(p-1)$, the number of terms J_i for lag i, and the cointegrating rank r.

The cointegrating rank can be estimated using the trace test as discussed in the last subsection. Alternatively, it can also be included in a joint information criterion such as SBIC or the PIC of Phillips and Ploberger [26]. With respect to the number of terms, the first approach will be to choose the same number in each lag initially. After a joint number has been found, we may search individually for reduction of the number of terms.

Often, the specification will be limited by the large number of parameters to be estimated. This puts strong restrictions on the number of lags/terms which may be included for typical sample sizes.

5.5. *Restrictions on cointegrating relations*

In (7) we did not impose any restriction on $\Pi = \alpha\beta'$. However, in some applications one may want to know whether there are (and if so, how many) cointegrating relations that

[4] For example: Stata uses critical values from Johansen [19], while E-Views, Stats models in Python and MATLAB rely on critical values from MacKinnon et al. [22].

(i) include a subset of variables in a subset of regions.

(ii) include cointegrating relations within a region that are identical for a number of regions.

Both types of restrictions can be formulated as $\beta = H\varphi$. For (i), H is a subset of columns of the identity matrix, where each omitted column represents one zero restriction. (ii) can be obtained by stacking the identity matrix (potentially padded with zeros for regions not to be included).

Using the format $(H_1\varphi_1, H_2\varphi_2)$, we can also impose the assumption that only some of the cointegrating relations follow this pattern. We can also combine the above cases leading to a flexible way to include different sources of cointegration into the model.

Since all of these restrictions are linear restrictions on the matrix β, the existence of such a structure can be tested in a standard fashion as explained in Johansen [19] (in section 7.2).

6. Numerical Simulations

In this section, we present a series of simulation experiments to evaluate the empirical performance of the proposed models. First, stationary $MaTS$ are investigated, and then the cointegrated $I(1)$ $MaTS$ case is examined.

6.1. Stationary MaTS

This section presents simulation experiments to show the efficacy of the proposed estimation methodologies for stationary $MaTS$. The simulation studies are organized into three parts. First, we evaluate the advantages of fitting a multi-term MAR model ($MAR_J(p)$) over a single-term MAR model ($MAR(p)$) for a simulated $MAR_J(p)$ process. We compare the estimation accuracy of the proposed estimators against that of single-term MAR estimators, as well as the VAR estimator for the vectorized process. Second, we investigate the consistency of model selection procedures with respect to both the autoregressive order p and the number of terms J, and we evaluate the out-of-sample predictive performance of the respective models.

In the first group of simulations, the matrix data Y_t are generated according to the multi-term MAR model in (2) with $p = 1$ and $J = 3$ according to Assumption 1. The coefficient matrices $A_{1,j}$ and $B_{1,j}$ for $j = 1, 2, 3$, which are generated randomly from the independent standard normal distribution, are normalized to $\|A_{1,j}\|_F = 1$ for $j = 1, 2, 3$ for identifiability. For each of the generated data, we restrict (via rejection sampling) the value of the spectral radius of the companion matrix $\rho(\mathbf{\Phi}) < 0.80$ to maintain the stability condition in Assumption 1.

The noise term is (i.i.d.) normal with covariance matrix $\Sigma_u = \mathrm{Var}(\mathrm{vec}(U_t))$. We consider two covariance structures for the noise process U_t:

- Case-I: The generic error covariance matrix without any structure: $\Sigma_u = P\Lambda P'$, where the elements of the diagonal matrix Λ are drawn by i.i.d. absolute standard normal random variables and P is an orthonormal random matrix of eigenvectors. This setting ensures that Σ_u is a symmetric positive definite matrix.

- Case-II: Separable covariance structure with Kronecker product form: $\Sigma_u = \Sigma_c \otimes \Sigma_r$, where Σ_c, column-wise covariance matrix, and Σ_r, row-wise covariance matrix, are generated analogously to Σ_u in Case-I.

Across all settings, we discard the first 1000 samples as a burn-in period from generated observations and repeat the simulations 100 times.

To evaluate the performance, we consider the following performance measures: The first is the comparison of estimated coefficients such that:

$$\text{Coefficient. error} = \sum_{r=1}^{p} log\|\widehat{\mathbf{\Phi}}_r - \mathbf{\Phi}_r\|_F^2, \quad p = 1 \tag{19}$$

where $\mathbf{\Phi}_r = \sum_{j_1=1}^{J_1}(B_{r,j_r} \otimes A_{r,j_r})$ are true values, $\widehat{\mathbf{\Phi}}_r = \sum_{j_1=1}^{J_1}(\widehat{B}_{r,j_r} \otimes \widehat{A}_{r,j_r})$ estimated values for multi-term MAR estimates with $J_1 = 3$, $\widehat{\mathbf{\Phi}}_r = (\widehat{B}_r \otimes \widehat{A}_r)$ estimated values for single-term MAR estimates and $\widehat{\mathbf{A}}_r$ for VAR estimates. In addition, we compare the forecast performance of estimations such as:

$$MSPE = \frac{1}{M \times N \times T_v} \sum_{h=1}^{T_v} e_{T+h} e'_{T+h} \tag{20}$$

where $\hat{Y}_{T+h|T}$ is an h-step-ahead forecast obtained with information up to the last sample size T for Y_{T+h}. T_v is the validation period. To simplify interpretation, we present the mean squared prediction error (MSPE) results relative to the benchmark model, which is unrestricted VAR, is indexed as 0 in the figures. i represents the corresponding estimator.

6.1.1. Estimation Accuracy

We compare three model structures ($VAR(p)$, $MAR(p)$ and $MAR_J(p)$) as well as three estimators for the $MAR(p)$ and $MAR_J(p)$ models ($ALSE$ for the alternating least squares estimator given in Algorithm 1, MLE with matrix normal distribution given in Algorithm S1 in the supplemental material and $qMLE$ discussed in Section 3.2). As numerical experiments yield similar results, we only report Case-I here, leaving the rest (Case-II) to supplementary material (Supplement S2). Figure 1 shows the simulation results in terms of the estimator error measures defined in (19) with Case-I covariance structure. We give the box-plot figures of the error of estimated coefficients defined above with the sample size increasing from the left column to the right with $T = \{100, 500, 1000\}$ and the rows correspond to increasing dimensions from top to bottom with $(M, N) = \{(2, 3), (3, 5), (6, 8)\}$. In each sub-figure, the VAR estimators are abbreviated as $VAR(p)$ in the first position, MAR estimators with single-term refer to $ALSE(p)$, $MLE(p)$ and $qMLE(p)$ in the following three positions, respectively, and multi-term MAR estimators refer to $ALSE_J(p)$, $MLE_J(p)$, $qMLE_J(p)$ in the last three positions. In these sub-figures, the lower values correspond to better estimation accuracy.

Figure 1 shows that both multi-term MAR estimators are superior to the others. Although $ALSE_J(p)$ and $MLE_J(p)$ are very similar in all setups, $MLE_J(p)$ performs slightly better. Interestingly, while $qMLE_J(p)$ is worse than the other two in small sample size with high complexity, it becomes better when the sample size gets larger. $qMLE_J(p)$ generally shows the best performance with the least bias and variance. The first row of the figure, which is an example of the small matrix size of M, N (i.e. $(2, 3)$), has the maximum effective term length of the multi-term MAR models. (i.e. $min(M^2, N^2) - 1$). In these situations, although there is not so much difference between using VAR or multi-term MAR in terms of number of parameter estimates, multi-term estimators still offer better performance.

The general results of these simulations including the supplementary tables are that all multi-term MAR estimators clearly outperform the stacked VAR estimator and single-term MAR estimators. Second, it can be observed that as the sample size and dimensions increase, the error of the estimation coefficient decreases accordingly, and multi-term $qMLE$ is consistently the most accurate and stable estimator.

Another expected result is that the misspecified MAR estimator shows a bias that does not vanish for larger sample sizes. The unrestricted VAR, on the other hand, demonstrates a large variance particularly in the higher dimensional cases. Consequently in our simulations, MAR beats VAR in small samples despite bias, while VAR gains accuracy relative to MAR in larger samples when the bias term dominates the variance.

In summary, MAR models clearly outperform VAR models when complexity is high and the number of observations is small. Overall, the results highlight the importance of sufficient data in achieving reliable estimation and emphasize the robustness and efficiency of multi-term MAR in more complex time series.

6.1.2. Forecasting performance including integer specification

Finally, we show the prediction performance of the corresponding models, where the lag length and the number of terms are determined by information criteria. For each model, we only show the information criterion with the best prediction performance. SBIC achieved the best result for $MAR_J(p)$ in general. Therefore, Figure 2 shows the forecast performance of the SBIC-selected models for $MAR_J(p)$ (red solid line) and the forecast performance of the best model for $MAR(p)$ (blue dotted line) and $VAR(p)$ (orange dashed line) , comparing across different sample sizes and model complexities, where the prediction mean square error (PMSE) is tracked over the forecast horizons. In all cases, $MAR_J(p)$ generally outperforms both $MAR(p)$ and $VAR(p)$, producing consistently lower PMSE, especially in larger samples and more complex models. The standard $MAR(p)$ model performs moderately well but is consistently less accurate than $MAR_J(p)$, while the $VAR(p)$ model exhibits the highest PMSE across horizons, especially in moderate and complex structures with small and medium sample size, reflecting over-parameterization and inefficiency. Overall, the results highlight the robustness of SBIC in selecting parsimonious yet accurate models, with $MAR_J(p)$ delivering the most reliable forecasts across varying dimensions and forecast horizons. Further discussion regarding model selection can be found in the specification accuracy section of the supplementary material.

In summary, Figure 2 highlights SBIC combined model selection appears to be the most reliable approach for both correct model identification and accurate forecasting, particularly in moderate and complex high-dimensional cases

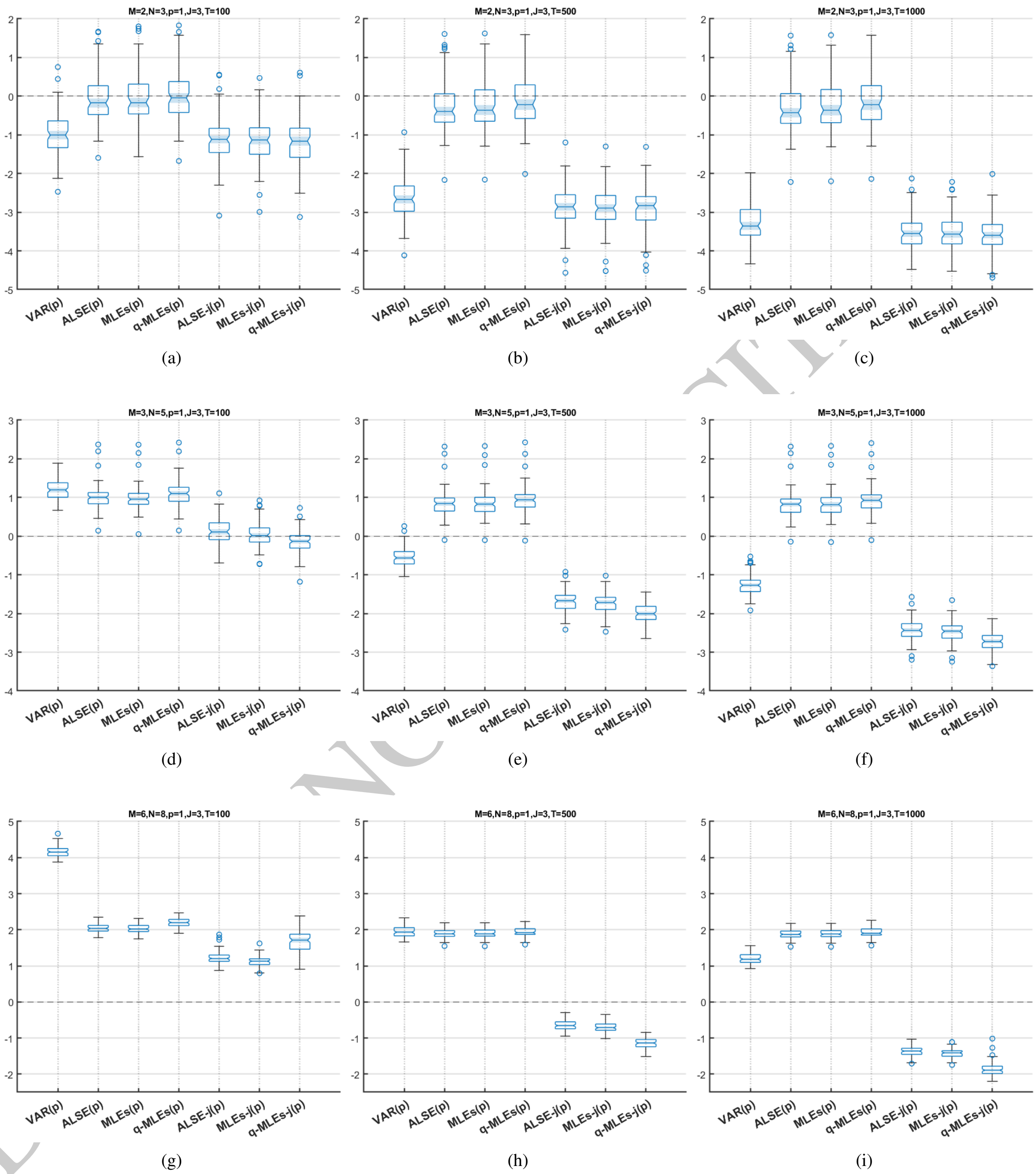

Figure 1: Comparison of estimators under Case-I. $VAR(p)$, $ALSE(p)$, $MLE(p)$, $qMLE(p)$, $ALSE_J(p)$, $MLE_J(p)$, $qMLE_J(p)$ (from left to right in each plot). Rows correspond to $(M, N) = \{(2, 3), (3, 5), (6, 8)\}$ and columns correspond to $T = \{100, 500, 1000\}$. Vertical axis are value of coefficient error.

and large-sample settings. However, note that in these simulations, we only considered the specification of a uniform number of terms for all lags. Given the high dimensionality it seems advisable to design specific selection procedures that allow to deviate from a uniform term number. This is left for future research.

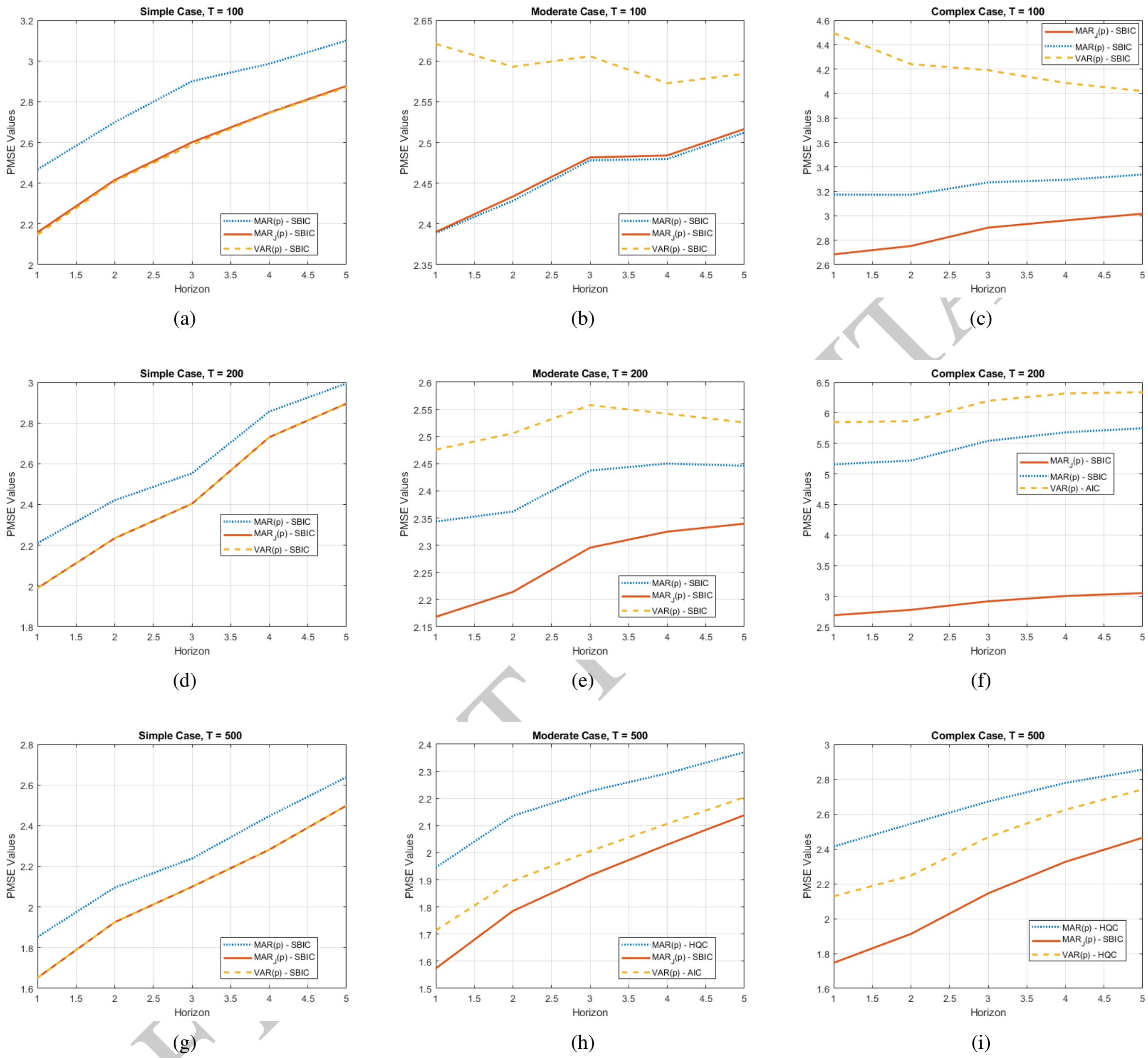

Figure 2: Forecast performance of selected models across different forecasting horizons for three data-generating scenarios (Small, Medium, and Large which have dimensions $(M, N) = \{(2, 3), (4, 5)(5, 10)\}$ respectively) under sample sizes $T = \{100, 200, 500\}$. Subfigures (a), (d), (g) correspond to the simple case; (b), (e), (h) correspond to the medium case; and (c), (f), (i) correspond to the large case. Each plot compares the mean square prediction error (MSPE) of three methods over given horizon: $MAR(p)$ (blue dashed line), $MAR_J(p)$ (red solid line), $VAR(p)$ (orange dashed line).

6.2. *Non-stationary MaTS*

This section covers simulation studies of the I(1)-cointegrated $MaTS$. We conducted a simulation study to investigate the performance of our rank selection method, which is discussed in Section 5.2 and Algorithm 3 for $vECMAR_J(p-1)$ model in equation (15). The data-generating process (DGP) is examined for the $vECMAR(2)$ model with $M = 4$ and $N = 5$ as defined in equation (15). The sample sizes are taken as $T = 200$ and $T = 500$ after the first 1000 taken as the burn-in period, and the simulations are repeated 100 times. We explore different cointegration rank settings in the models: (i) no cointegration ($r = 0$), and (ii) existence of cointegration ($r = 1, 6, 11, 16, 19$).

Given the true rank r, the coefficient matrix, $\alpha\beta' = U\Lambda V'$, is constructed randomly by first generating two orthonormal matrices U, V with dimensions $MN \times r$ by using the QR decomposition from matrices with each element drawn from a standard normal distribution. Then the elements of $r \times r$ diagonal matrix Λ are drawn i.i.d. as the absolute value of standard normal random variables. For the short-term coefficients, row and column $\Gamma_{k,i} = Q_1 \Lambda Q_2'$, are generated in the same way except that Q_1, Q_2 and Λ, $M \times M$ for the row matrix and $N \times N$ for the column matrix. We ensure that the stability condition holds, $\rho(\mathbf{\Gamma}) < 0.95$, for the companion matrix of $vECMAR_J(p-1)$. [5] The noise term is i.i.d. normal with the covariance matrix $\Sigma_u = Cov(vec(U_t))$ defined as the first case of the stationary section.

To evaluate the performance of the rank selection and estimation procedure, we calculate the average rank selected by the method, its corresponding standard deviation, and the frequency of correctly chosen ranks. In addition, we compare our model to the standard Johansens's $VECM$ estimation and we show the box plots of the estimator error:

$$E.error_i = log(\|\widehat{\mathbf{P}}_{\hat{\beta}_i} - \mathbf{P}_\beta\|_F^2) \tag{21}$$

where $\mathbf{P}_{\beta_i} = \beta_i(\beta_i'\beta_i)^{-1}\beta_i'$ is the projection matrix of β_i for the corresponding method and given β matrix.

True Rank	Method	T	Min. Rank	Median Rank	Max. Rank	Ave. Rank	S.D.	Success (%)
0	EC-MAR	(200)	0	0	0	0.00	0.00	100 %
	VECM	(200)	3	8	18	8.13	2.65	0 %
	EC-MAR	(500)	0	0	0	0.00	0.00	100 %
	VECM	(500)	0	1	5	1.44	0.21	21 %
1	EC-MAR	(200)	0	0	1	0.42	0.43	42 %
	VECM	(200)	4	9	19	9.21	3.19	0 %
	EC-MAR	(500)	0	1	1	0.86	0.34	86 %
	VECM	(500)	1	2	8	2.40	1.18	23 %
6	EC-MAR	(200)	1	4	6	4.19	0.90	6 %
	VECM	(200)	6	9	16	9.72	2.52	9 %
	EC-MAR	(500)	4	6	6	5.66	0.53	69 %
	VECM	(500)	5	6	10	6.63	0.99	49 %
11	EC-MAR	(200)	6	9	12	9.21	1.20	11 %
	VECM	(200)	9	12	20	11.82	1.90	26 %
	EC-MAR	(500)	10	11	14	10.86	0.63	69 %
	VECM	(500)	10	11	20	11.23	1.12	72 %
16	EC-MAR	(200)	12	15	19	15.18	1.21	31 %
	VECM	(200)	10	15	19	14.96	1.51	17 %
	EC-MAR	(500)	14	16	17	15.96	0.37	89 %
	VECM	(500)	14	16	17	15.95	0.43	84 %
19	EC-MAR	(200)	14	19	20	18.55	0.77	60 %
	VECM	(200)	14	19	20	18.40	0.88	47 %
	EC-MAR	(500)	19	19	20	19.02	0.14	98 %
	VECM	(500)	18	19	20	19.05	0.26	93 %

Table 1: Comparison of vECMAR and VECM estimations with cointegration rank selection for dimensions $M = 4$ and $N = 5$ data under covariance structure is case-I. Simulation results are from $T = 200$ and $T = 500$ with no cointegration ($r = 0$) and 5 rank cases ($r = 1, 6, 11, 16, 19$). The last column shows the success rate over 100 repetitions. The critical values in the third column of Table A.2 have been used.

Table 1 presents a comparative analysis of the $vECMAR$ and $VECM$ estimation methods in determining cointegration ranks under various true rank conditions and sample sizes ($T = 200$ and $T = 500$) for data with dimensions $M = 4$ and $N = 5$ under covariance structure Case-I. Performance metrics include minimum, median, maximum,

[5] To ensure that the system is a pure I(1) process, all individual series are additionally checked with the ADF unit root test.

average and standard deviation of the estimated ranks, along with the success rate over 100 replications. The results reveal that $vECMAR$ consistently outperforms $VECM$ across nearly all rank scenarios, particularly as the sample size increases. When the true rank is zero (no cointegration), $vECMAR$ achieves perfect accuracy for both sample sizes, whereas $VECM$ frequently overestimates the rank. For non-zero ranks, $vECMAR$ maintains lower average rank errors and higher success rates, especially at higher sample sizes, whereas $VECM$ exhibits larger deviations and lower success rates. The advantage of $vECMAR$ becomes particularly evident in higher-rank cases (e.g., ranks 16 and 19), where it achieves near-perfect identification accuracy with minimal variability. Overall, the findings suggest that $vECMAR$ provides more reliable and precise estimation of the cointegration ranks than traditional $VECM$, especially in larger samples and more complex systems.

Figure 3 presents box-plots comparing the estimation accuracy of the space spanned by the columns of the estimated β matrices (defined in equation- (21)) for the same estimation results as used for Table 1, obtained using the $vECMAR$ and $VECM$ methods across different true cointegration ranks ($r = 1, 6, 11, 16$) and sample sizes ($T = 200$ and $T = 500$). The plots reveal clear performance differences between the two estimators. For smaller samples ($T = 200$), $vECMAR$ consistently produces tighter distributions with fewer outliers, indicating greater stability and lower estimation variability compared to $VECM$. As the sample size increases to $T = 500$, both methods improve in precision, but $vECMAR$ maintains a higher concentration around the central values, demonstrating its robustness and efficiency even in larger-sample settings. The $VECM$ method exhibits wider interquartile ranges and more extreme outliers, particularly for lower ranks ($r = 1$ and $r = 6$), suggesting greater sensitivity to sample fluctuations. Overall, the figure illustrates that $vECMAR$ yields more accurate and consistent projections of the cointegration space across all rank scenarios, reaffirming its advantage over the traditional $VECM$ approach.

Taken together, the results of Table 1 and Figure 3 clearly suggest that $vECMAR$ provides a more accurate, stable and sample-efficient estimation of both the cointegration rank and the corresponding β matrices compared with $VECM$. This advantage becomes more pronounced with increasing sample size, confirming the reliability of $vECMAR$ for cointegration analysis.

We also compare the two models when considering the size adjusted critical values presented in the Table A.2 for determining the cointegration rank. In this case, although both models produce better results than the results we obtain (in Table 1) when using asymptotic critical values, $vECMAR$ still outperforms $VECM$. The results are presented in Table-S2, which can be found in the supplementary material.

7. Conclusions and Discussions

This study examined matrix autoregressive (MAR) models for both stationary and cointegrated ($I(1)$) processes. We introduced the multi-term matrix autoregressive model, $MAR_J(p)$, together with an identification strategy based on QR decomposition. Three estimation procedures—ALSE, MLE (under the matrix normal distribution), and qMLE—were proposed and evaluated. Simulation results demonstrate that these estimators outperform traditional VAR and single-term MAR models in both in-sample estimation and out-of-sample forecasting when the data-generating process follows a multi-term MAR structure. Among these methods, qMLE consistently achieved the best performance, especially in high-dimensional and complex systems, and is therefore recommended for empirical use.

The model specification was assessed using standard information criteria—AIC, SBIC and HQC. Our findings indicate that SBIC provides the most reliable identification of lag and term lengths. Nevertheless, given the intermediate nature of the multi-term model between MAR and VAR representations, there remains scope for developing more refined selection procedures to ensure accurate model identification and improved forecasting reliability.

For the integrated case, we showed that the one-term MAR model in levels transforms naturally into the multi-term MAR form upon integration, and a similar relationship holds between the one-term $MECM$ and multi term MAR models in levels. We further demonstrated that imposing Kronecker product restrictions on the error-correction matrix Π is inconsistent with the standard cointegrated VAR framework. Consequently, we propose an unrestricted specification of Π, since a sum of Kronecker products cannot, in general, be represented as a single Kronecker product or any of its special cases that coincide with cointegration framework. This unrestricted formulation preserves the theoretical validity of the cointegrated VAR framework while retaining the matrix structure of the data. Although it limits the direct interpretation of row- and column-wise interactions within Π, the approach remains a theoretically coherent and practically tractable foundation for further development.

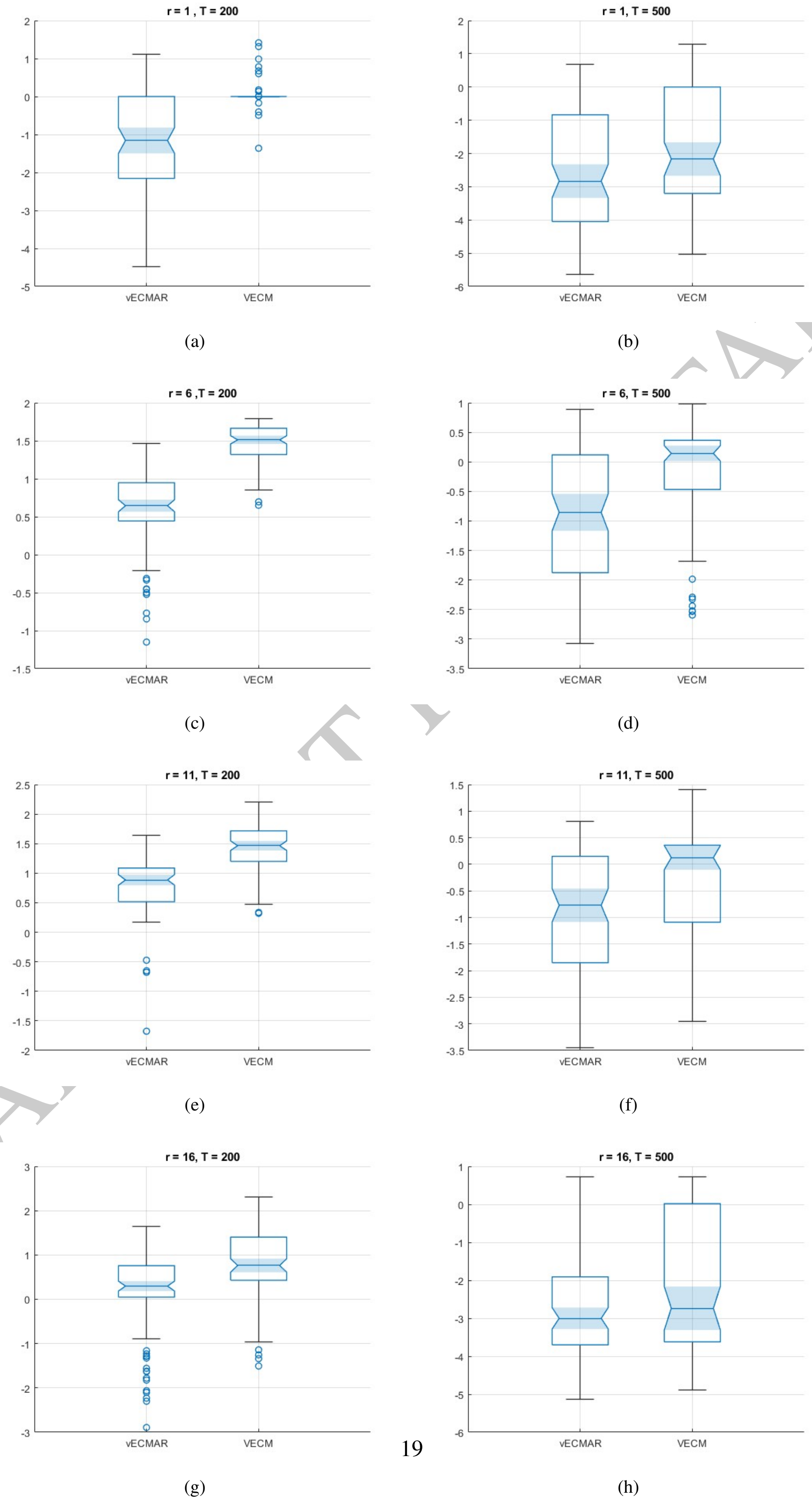

Figure 3: Comparison of projection matrix of estimated Π matrices.

We also proposed a *vECMAR* model which uses a Johansen-type trace test to determine the cointegration rank within the *MAR* framework and derived extended critical values applicable to higher-dimensional systems. These expanded values address an important gap in the literature, enabling reliable rank testing and inference in matrix-valued time series models of previously infeasible dimensions.

Future work could extend the proposed framework in several promising directions. First, developing model selection criteria specifically tailored to the multi-term MAR structure would enhance the accuracy of term and lag identification. Second, reinvestigating the critical values of trace statistics for high dimensions. Finally, extending the multi-term MAR framework to structural-break, or regime-switching environments would broaden its applicability to complex empirical domains such as image processing, financial networks, macroeconomic panels, and spatio-temporal econometric systems.

8. Acknowledgements

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Appendix A. Critical Values of Trace Test for Cointegration

This section presents the calculated critical values for the trace statistics used in the simulation experiments Table A.2 reports the 95% critical values for both the asymptotic distribution and the model specific sample size-adjusted distributions. For comparison purposes, it also includes the reference values from MacKinnon et al. [22] et al. In addition, in the supplementary material, Table-S3 provides extended critical values by reporting additional asymptotic percentiles—namely 90%, 95%, 97.5% and 99%.

Table A.2: 5.0% Critical values for trace test for cointegration rank without constant and deterministics.

p-r	MacKinnon	Infinite	VECM T=200	vECMAR T=200	VECM T=500	vECMAR T=500
1	4.13	4.13	3.99	2.91	4.48	3.95
2	12.32	12.26	12.11	12.49	13.00	13.16
3	24.27	24.59	25.34	26.85	26.72	25.67
4	40.17	40.36	45.49	47.82	40.94	40.74
5	60.06	60.20	64.42	63.68	64.53	62.28
6	83.94	83.69	91.14	93.14	88.98	89.10
7	111.78	111.86	119.31	117.71	116.78	118.77
8	143.67	144.41	164.36	154.74	157.40	158.18
9	179.50	179.46	206.35	195.30	198.32	191.49
10	219.39	220.18	249.74	234.92	242.53	236.34
11	263.26	263.65	308.81	259.51	296.28	273.48
12	311.12	311.60	363.80	303.18	353.46	316.90
13		362.92	431.52	352.32	397.66	366.24
14		419.10	523.51	379.56	455.97	428.75
15		479.40	602.27	425.07	536.95	481.10
16		542.61	685.73	456.60	594.47	491.18
17		612.15	785.00	505.64	672.60	541.05
18		683.42	897.49	581.06	764.49	609.41
19		761.20	1021.17	609.59	857.24	633.77
20		839.09	1169.28	671.86	953.50	707.58

The asymptotic distributions of the Trace and Maximal Eigenvalue statistics are given, respectively, by the trace and maximal eigenvalue of

$$\int_0^1 dB(u)\, F(u)' \left(\int_0^1 F(u)\, F(u)' du \right)^{-1} \int_0^1 F(u)\, (dB(u))'$$

where $u \in [0, 1]$, $B(u)$ denotes a $(k-r)$ dimensional Brownian motion on the unit interval, and $F(u)$ contains functions of $(k-r)$ dimensional Brownian motion which is approximated using a Gaussian random walk with $N(0, I)$ innovations. (For more details Johansen [19]). The limit distribution above can be simulated via Monte Carlo methods by generating a k-dimensional random walk with $(k-r)$ stochastic common trend components. We then compute the trace statistic, based on the likelihood ratio test from the reduced-rank regression framework. Using a sufficiently large sample size, $T = 5000$, we performed $10,000$ replications for each $k-r = 1, \ldots, 20$ and computed the empirical quantiles.

To estimate the size-adjusted critical values, we generated 4×5 dimensional MaTS datasets for each rank specification $r = 0, \ldots, 19$ and for sample size $T = 200, 500$, as described in Section 6.2. For each pair of (T, r), we replicated the simulations 100 times. We then separated the calculated eigenvalues for each model using the proposed estimation method and computed the empirical quantiles across the replications. As a note, we used the asymptotic critical values for the reduced estimation of initial cointegration rank for $vECMAR$.

Critical values for the maximal eigenvalue test can be obtained in a similar fashion and are omitted here for brevity.

Appendix B. Proofs

Proof of Theorem 1. (I) The criterion function $ll_T(\theta)$ includes the term $\Sigma_u(\theta)$ and $\frac{1}{T}\sum_{t=p+1}^{T} \hat{u}_t(\theta)\hat{u}_t(\theta)'$ where

$$\hat{u}_t(\theta) = \mathrm{vec}(\hat{U}_t(\theta)) = \mathrm{vec}\left(Y_t - \sum_{j=1}^{p} \left(\sum_{r=1}^{J_j} A_{j,r}(\theta) Y_{t-j} B_{j,r}(\theta)'\right) \right) = y_t - \sum_{j=1}^{p} \left(\sum_{r=1}^{J_j} (B_{j,r}(\theta) \otimes A_{j,r}(\theta) \right) y_{t-j}.$$

Thus,

$$\frac{1}{T} \sum_{t=p+1}^{T} \hat{u}_t(\theta)\hat{u}_t(\theta)' = \sum_{i=0}^{p} \sum_{j=0}^{p} \mathcal{A}_i(\theta) \left[\frac{1}{T} \sum_{t=p+1}^{T} y_{t-i} y_{t-j}' \right] \mathcal{A}_j(\theta)'$$

where $\mathcal{A}_0(\theta) = I_{MN}, \mathcal{A}_j(\theta) = -\sum_{r=1}^{J_j}(B_{j,r}(\theta) \otimes A_{j,r}(\theta)), j = 1, ..., p$. Under the assumptions of the theorem $\frac{1}{T}\sum_{t=p+1}^{T} y_{t-i} y_{t-j}' \to \mathbb{E} y_{t-i+j} y_t' =: C_{i-j}$ (say) for all $0 \le i, j \le p$. The proof of Theorem A.1 of Johansen [27] implies that $\|\mathcal{A}_j(\hat{\theta}) - \mathcal{A}_j(\theta_\circ)\| \to 0$. Similarly one can show that $\hat{\theta}$ (for large enough T almost surely) lies in the subset such that $\lambda_{|min|}(\Sigma_u(\theta)) \ge \underline{c}, \lambda_{|max|}(\Sigma_u(\theta)) \le \overline{C}$ for suitable constants $0 < \underline{c} < \overline{C}$. It follows that asymptotically with probability tending to one $\hat{\theta}$ is such that $\mathcal{A}_j(\hat{\theta})$ and $\Sigma_u(\hat{\theta})$ lies in a compact neighborhood of the true matrices. Since the parameterization is continuous at $\theta_\circ$ the corresponding set of vectors θ also is compact. Furthermore, uniformly on this compact set,

$$ll_T(\theta) \to \log|\Sigma_u(\theta)| + \mathrm{tr}\left[\left(\sum_{i=0}^{p} \sum_{j=0}^{p} \mathcal{A}_i(\theta) C_{i-j} \mathcal{A}_j(\theta)' \right) \Sigma_u(\theta)^{-1} \right].$$

This function is minimized at $\theta_\circ$. This implies consistency for the estimator $\hat{\theta}$.

(II) With respect to asymptotic normality, the arguments from above can be used noting that

$$\sqrt{T}\mathrm{vec}(\frac{1}{T} \sum_{t=p+1}^{T} y_t y_{t-j}' - C_j)$$

is asymptotically normal (also jointly for $j = 0, 1, ..., p$) under the assumptions of the theorem (see, for example, the arguments around Lemma 4.3.4 of Hannan and Deistler [15]). Then the result follows from the usual application of a Taylor expansion of the first order conditions in conjunction with the differentiability of the mapping $\varphi(\theta)$. □

Proof of Theorem 2. (I) Consider (11) and transform using T_A and T_B:

$$T_A^{-1} Y_t (T_B^{-1})' = \tilde{Y}_t = \begin{pmatrix} I_{n_A} & 0 \\ 0 & A_{1,1,\bullet} \end{pmatrix} \tilde{Y}_{t-1} \begin{pmatrix} I_{n_B} & 0 \\ 0 & B'_{1,1,\bullet} \end{pmatrix} + \tilde{U}_t.$$

Vectorising we obtain

$$\tilde{y}_t = \left(\begin{pmatrix} I_{n_B} & 0 \\ 0 & B_{1,1,\bullet} \end{pmatrix} \otimes \begin{pmatrix} I_{n_A} & 0 \\ 0 & A_{1,1,\bullet} \end{pmatrix} \right) \tilde{y}_{t-1} + \tilde{u}_t.$$

It can be readily seen that the eigenvalues of the Kronecker product equal the product of the eigenvalues of the matrices in the product. Since n_A and n_B of the eigenvalues equal $z = 1$ respectively, we obtain $n_A n_B$ products equal to 1. All remaining products have magnitude smaller than one. This shows (I).

(II) Vectorising we get

$$\Delta y_t = (\alpha_c \beta_c' \otimes \alpha_r \beta_r') y_{t-1} + u_t = (\alpha_c \otimes \alpha_r)(\beta_c' \otimes \beta_r') y_{t-1} + u_t$$

We obtain similarly that the eigenvalues of $\Pi = (\alpha_c \beta_c' \otimes \alpha_r \beta_r')$ are the products of the eigenvalues of $\alpha_r \beta_r'$ and of $\alpha_c \beta_c'$. In the error correction formulation we are interested in the zero eigenvalues. The product is zero, if at least one of its components is zero. In order to obtain non-zero terms hence both factors need to be non-zero. This happens in $r_r r_c$ such products since, for example, the matrix $\alpha_r \beta_r'$ has r_r nonzero eigenvalues and $M - r_r$ zero ones. □

Proof of Theorem 3. The proof uses results in the Appendix of Johansen [27]. Theorem A.1 of Johansen [27], there states consistency of the coefficient matrices in $I(2)$ settings. The idea of the proof also holds in the $I(1)$ case (compare [27], p. 458, below the proof of Theorem A.2) in VECM formulation using the model equation

$$\Delta y_t = \alpha(\theta)\beta(\theta)' y_{t-1} + \sum_{k=1}^{p-1} \mathcal{G}_j(\theta) \Delta y_{t-k} + u_t. \tag{B.1}$$

Bijectivity and continuity at $\theta_\circ$ of the parameterisation φ and its inverse implies that for each $\epsilon > 0, \delta > 0$ there exists an ϵ-neighborhood $N(\alpha_\circ, \beta_\circ, \mathcal{G}_{j,\circ}; \epsilon)$ such that $\theta \in N(\theta_\circ, \delta)$ is in the δ–neighborhood of $\theta_\circ$, if $\varphi(\theta) \in N(\alpha_\circ, \beta_\circ, \mathcal{G}_{j,\circ}; \epsilon)$. [27] Theorem A.1 then shows that the unrestricted estimators are in the neighborhood $N(\alpha_\circ, \beta_\circ, \mathcal{G}_{j,\circ}; \epsilon)$ which hence implies that the estimate $\hat{\theta}$ must be in the neighborhood $N(\theta_\circ, \delta)$. Note that in our situation also parameters for $\Sigma_u(\theta)$ are contained in the parameter vector, whereas [27] uses an unrestricted matrix Σ_u. We assume that Θ is the product space between parameter vectors σ parameterising $\Sigma_u(\theta)$ and the remaining parameters. As in the proof of Theorem 1 one can verify that the eigenvalues for $\Sigma_u(\theta)$ fulfill lower and upper bounds for the estimator asymptotically with probability tending to one. Thus the estimator lies in a closed neighborhood of $\theta_\circ$ for T large enough with probability tending to one. Then again Theorem A.1 of Johansen [27] implies consistency $\hat{\theta} \to \theta_\circ$. Furthermore, the parameters $\theta_{\beta,L}$ corresponding to the column space of β converge faster than $1/\sqrt{T}$ using the arguments of the proof of Theorem A.1 of Johansen [27].

(II) and (III) then follow from the arguments in the proof of Lemma 1 of Johansen [27]: Define $B_\circ = [\beta_\circ, \beta_\perp] \in \mathbb{R}^{MN \times MN}$ such that $B_\circ' B_\circ = I_{MN}$ for some matrix $\beta_\perp$. Further, let $B_\circ' y_t = (y_{t,\beta}', y_{t,\perp}')'$ such that $(y_{t,\beta})_{t\in\mathbb{Z}}$ is stationary and $(y_{t,\perp})_{t\in\mathbb{Z}}$ is $I(1)$. Define the reparameterization $\alpha(\theta)\beta(\theta)' = \alpha(\tilde{\theta})\beta(\tilde{\theta})'$ using $\beta(\tilde{\theta}) = \beta_\circ + \beta_\perp \delta_\beta, \delta_\beta \in \mathbb{R}^{(NM-r)\times r}$ such that $\beta(\tilde{\theta})'\beta_\circ = I_r$. We first derive the asymptotic distribution for this parameterization. Hence δ_β replaces $\theta_{\beta,L}$. Note that all entries of δ_β are free parameters. The parameter vectors for α and $\theta_{\beta,R}$ adjust accordingly. Subsequently, the change of coordinates will be discussed. Then

$$\begin{aligned} u_t(\tilde{\theta}) &= \Delta y_t - \alpha(\tilde{\theta})\beta(\tilde{\theta})'(\beta_\circ y_{t-1,\beta} + \beta_\perp y_{t-1,\perp}) + \sum_{k=1}^{p-1} \mathcal{G}_j(\theta) \Delta y_{t-k} \\ &= u_t + \alpha(\tilde{\theta})(\beta_\circ - \beta(\tilde{\theta}))' y_{t-1} + (\alpha_\circ - \alpha(\tilde{\theta})) y_{t-1,\beta} + \sum_{k=1}^{p-1} (\mathcal{G}_j(\theta_\circ) - \mathcal{G}_j(\theta)) \Delta y_{t-k} \\ &= u_t + \alpha_\circ \delta_\beta' y_{t-1,\perp} + (\alpha(\tilde{\theta}) - \alpha_\circ)\delta_\beta' y_{t-1,\perp} + (\alpha_\circ - \alpha(\tilde{\theta})) y_{t-1,\beta} + \sum_{k=1}^{p-1} (\mathcal{G}_j(\theta_\circ) - \mathcal{G}_j(\theta)) \Delta y_{t-k}. \end{aligned}$$

Due to consistency and the fact that the true parameter vector is an interior point of the parameter set, the estimator maximizing the Gaussian likelihood is given as the solution to the first order equations. These involve the partial derivatives $\partial_a \hat{u}_t(\tilde{\theta})$ with respect to all components of $\tilde{\theta}$. The parameter vector $\tilde{\theta}$ can be partitioned into three parts: the first part, δ_β, parameterises the column space of β. The second part $\tilde{\theta}_\bullet$ includes $\theta_{\beta,R}$, parameters for α and $\mathcal{G}_j$. Finally, the third part equals σ, parameters for $\Sigma_u(\theta)$.

The equation above implies that the derivative with respect to δ_β involves the $I(1)$ process $y_{t-1,\perp}$. The derivative with respect to any other parameter component results in a stationary process (plus for $\alpha(\tilde{\theta})$ a contribution from $\delta_\beta' y_{t-1,\perp}$ which due to $\sqrt{T}\|\delta_\beta\| \to 0$ from the last statement in the proof of (I) is negligible).

Therefore as the proof of Lemma 1 in the Appendix of [27] we get the first order conditions for the first two parts and the third part:

$$0 = \text{tr}\left[\frac{1}{T}\sum_{t=p+1}^{T} \partial_a \hat{u}_t(\tilde{\theta})\hat{u}_t(\tilde{\theta})'\Sigma_u(\tilde{\theta})^{-1}\right] = \text{tr}\left[\left(\frac{1}{T}\sum_{t=p+1}^{T} \partial_a \hat{u}_t(\tilde{\theta})u_t' + \frac{1}{T}\sum_{t=p+1}^{T} \partial_a \hat{u}_t(\tilde{\theta})(\hat{u}_t(\tilde{\theta}) - u_t)'\right)\Sigma_u(\tilde{\theta})^{-1}\right],$$

$$0 = \partial_\sigma ll_T(\theta) = \text{tr}\left[\Sigma_u(\sigma)^{-1}\partial_\sigma \Sigma_u(\sigma) - \Sigma_u(\sigma)^{-1}\partial_\sigma \Sigma_u(\sigma)\Sigma_u(\sigma)^{-1}\frac{1}{T}\sum_{p+1}^{T}\hat{u}_t(\tilde{\theta})\hat{u}_t(\tilde{\theta})'\right].$$

From above it follows that

$$\hat{u}_t(\tilde{\theta}) - u_t = \frac{\partial_\beta \hat{u}_t(\tilde{\theta})}{T} T \text{vec}(\delta_\beta) + \frac{\partial_\bullet \hat{u}_t(\tilde{\theta})}{\sqrt{T}}\sqrt{T}(\tilde{\theta}_\bullet - \beta_{\bullet,\circ}) + R_T(\tilde{\theta}).$$

The linearization error $R_T(\tilde{\theta})$ is of lower order. Also the consistency result of (I) implies that in all occurrences the estimators $\tilde{\theta}$ can be replaced by its limit adding only negligible terms.

For δ_β note that all entries are free parameters. Then deriving $\hat{u}_t(\tilde{\theta})$ with respect to the (i,j) entry of δ_β equals $\alpha_\circ e_j e_i' y_{t-1,\perp}$, where e_i denotes the i-th vector of the canonical basis (with sloppy notation using the symbol for different dimensions of the vector). It follows that the corresponding derivative of $ll_T(\tilde{\theta})$ equals

$$\text{tr}\left[\frac{1}{T}\sum_{t=p+1}^{T}\alpha_\circ e_j e_i' y_{t-1,\perp}\hat{u}_t(\tilde{\theta})'\Sigma_u(\tilde{\theta})^{-1}\right] = e_i'\left(\frac{1}{T}\sum_{t=p+1}^{T} y_{t-1,\perp}\hat{u}_t(\tilde{\theta})'\Sigma_u(\tilde{\theta})^{-1}\alpha_\circ\right)e_j.$$

Deriving this again with respect to the (k,l) entry of δ_β one obtains

$$e_i'\left(\frac{1}{T}\sum_{t=p+1}^{T} y_{t-1,\perp}y_{t-1,\perp}'\right)e_k\left(e_l'\alpha_\circ'\Sigma_u(\tilde{\theta})^{-1}\alpha_\circ e_j\right).$$

From this it follows that the first order conditions for δ_β stacked with the first order conditions for $\tilde{\theta}_\bullet$ and $\hat{\sigma}$ multiplied with $\sqrt{T}$ converges to

$$\begin{pmatrix} \text{vec}\left[\int_0^1 W(r)dB(r)'\Sigma_u^{-1}\alpha_\circ'\right] \\ Z_\bullet \\ Z_\sigma \end{pmatrix} = \begin{pmatrix} H_{\beta,\beta} & 0 & 0 \\ 0 & H_{\bullet,\bullet} & 0 \\ 0 & 0 & H_{\sigma,\sigma}\end{pmatrix}\begin{pmatrix} T(\tilde{\theta}_\beta - \beta_\circ) \\ \sqrt{T}(\tilde{\theta}_\bullet - \beta_{\bullet,\circ}) \\ \sqrt{T}(\hat{\sigma} - \sigma_\circ)\end{pmatrix} + o_P(1),$$

$$H_{\beta,\beta} = -\alpha_\circ'\Sigma_u^{-1}\alpha_\circ \otimes \int_0^1 W(r)W(r)'dr,$$

$$H_{\bullet,\bullet} = -\left[\text{tr}\left(\Sigma_u^{-1}\mathbb{E}\partial_{\bullet,a}\hat{u}_t(\theta_\circ)\partial_{\bullet,b}\hat{u}_t(\theta_\circ)'\right)\right]_{a,b},$$

$$H_{\sigma,\sigma} = -\left[\text{tr}\left(\Sigma_u^{-1}\partial_{\sigma_a}\Sigma_u(\sigma)\Sigma_u^{-1}\partial_{\sigma_b}\Sigma_u(\sigma)]\right)\right]_{a,b}.$$

Here $W(r)$ denotes the Brownian motion related to $y_{t-1,\perp} \in \mathbb{R}^{NM-r}$ and the Brownian motion $B(r) \in \mathbb{R}^{NM}$ relates to u_t. These two Brownian motion are related according to $W(u) = C\alpha_\perp B(u)$ for some invertible matrix $C \in \mathbb{R}^{(MN-r)\times(MN-r)}$. It follows that the two Brownian motions appearing on the left hand side are uncorrelated.

$Z_\bullet$ denotes the Gaussian random limit of $\frac{1}{\sqrt{T}}\sum_{t=p+1}^{T}\partial_\bullet \hat{u}_t(\tilde{\theta})u_t'$. Finally Z_σ denotes the limiting Gaussian random variable corresponding to $[-\mathrm{tr}[\Sigma_u^{-1}\partial_{\sigma,a}\Sigma_u(\sigma)\Sigma_u^{-1}\frac{1}{\sqrt{T}}\sum_{t=p+1}^{T}(u_tu_t'-\Sigma_u)]]_a$.

From the first order conditions we also obtain $T^\gamma\delta_\beta \xrightarrow{p} 0$ for every $0<\gamma<1$. Normalizing β to an orthonormal column using $\hat{\beta} := \beta(\tilde{\theta})(\beta(\tilde{\theta})'\beta(\tilde{\theta}))^{-1/2}$ one obtains

$$\beta(\tilde{\theta})'\beta(\tilde{\theta}) = I_r + \delta_\beta'\delta_\beta = I_r + o_P(T^{-2\gamma}) = I_r + o_P(T^{-1}).$$

Then the differentiability of the Cholesky decomposition implies that $(\beta(\tilde{\theta})'\beta(\tilde{\theta}))^{-1/2} = I_r + o_P(T^{-1})$ such that we obtain $\hat{\beta}-\beta_\circ = \beta(\tilde{\theta})-\beta_\circ + o_P(T^{-1})$. Due to the differentiability of the QR decomposition, we obtain from the Delta method that the asymptotic distribution for the parameter vector $\theta_{\beta,L}$ equals the distribution derived above where the definition of $W(r)$ is changed by pre-mulitplication with the Jacobian matrix corresponding to the inverse of the mapping attaching $\beta(\theta)$ to $\theta_{\beta,L}$. Thus the asymptotic distribution of the parameter vector $\theta_{\beta,L}$ is mixed normal.

Asymptotic normality for the remaining parameters follows as usual due to the block diagonality established above. □

Appendix C. Supplementary data

Replication materials for simulations and estimation are available at https://github.com/kurtuluskidik/SEMaTS.git. The supplementary material includes additional results from simulations to support the findings in the main text.

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